Future Proves Past: Detecting Hidden Fraud under Imperfect Enforcement Labels

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Abstract

SEC enforcement labels are selective and incomplete proxies for the latent fraud they are meant to measure. This mismatch creates two problems: a model trained on them may learn the enforcement process rather than the fraud process, and accuracy measured against them cannot reveal whether a model detects the fraud enforcement misses. We address both with a *future-proves-past* design. Because accounting distortions eventually reverse, future accounting outcomes modestly refine the training signal into soft labels; we train a convolutional neural network on multi-year accounting panels with these labels and validate it entirely outside the enforcement set. The model gives up little SEC-labeled detection performance. Out of sample over 2000–2019, flagged firms complete seasoned equity offerings at nearly three times the base rate and earn large negative post-issuance abnormal returns, a long-short portfolio earns significant factor-adjusted alpha, and externally screened populations (post-scrutiny Arthur Andersen clients and bank-monitored borrowers) are systematically avoided. An identically specified network trained on the enforcement label, and benchmarks designed to predict it, reproduce little of this profile beyond a size-controlled tilt away from bank borrowers: the training target drives the difference. The results show that selective enforcement data can be repurposed, at modest cost, to reach fraud that enforcement itself never surfaces, and they caution that detection models should be judged by validation outside the very label on which they were trained.

Keywords: financial statement fraud; hidden fraud; SEC enforcement; weak supervision; deep learning

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1 Introduction

Financial statement fraud threatens market integrity, investor confidence, and the efficient allocation of capital (Lie, 2025). A large accounting literature has developed methods to detect fraud and earnings manipulation using accounting information, ranging from ratio-based screening models such as the M-score and F-score (Beneish, 1999; Dechow et al., 2011) to more recent machine-learning approaches that incorporate financial statements, textual disclosures, and ensemble methods (Kirkos et al., 2007; Ravisankar et al., 2011; Larcker and Zakolyukina, 2012; Brown et al., 2020; Bao et al., 2020). A common feature of this literature is the use of SEC enforcement actions, particularly Accounting and Auditing Enforcement Releases (AAERs), as the labels that stand in for actual fraud in model training and evaluation; surveying top-journal fraud research, Donelson et al. (2021) find that a majority of studies measure fraud with SEC cases alone.

However, recent evidence from Dyck et al. (2024) suggests that corporate fraud is substantially more pervasive than implied by observed SEC enforcement. Enforcement is constrained by institutional resources and legal standards, and its selection is systematic rather than random: investigations depend on factors such as geographic proximity to SEC offices (Kedia and Rajgopal, 2011), political connections (Correia, 2014), and private-sector scrutiny, salient public events, and SEC personnel incentives (Holzman et al., 2024). Consequently, AAER-based datasets contain systematic label noise, as the absence of an AAER may reflect enforcement constraints rather than the absence of fraudulent or questionable reporting. This concern is well documented. Karpoff et al. (2017) show that research conclusions depend on the misconduct database employed, while Donelson et al. (2021) demonstrate that relying on a single enforcement regime materially changes empirical inferences and recommend broadening fraud measures beyond AAERs. More broadly, the objective of fraud detection is not merely to predict observed enforcement outcomes, but to identify firms exhibiting fraud-like characteristics that warrant closer scrutiny. Models capable of recovering this latent signal can help regulators prioritize investigative resources and provide investors and auditors with earlier warnings of potential financial reporting misconduct.

In this paper, we distinguish the latent economic object of interest from the observed enforcement label. Following Dechow et al. (2011), we define *fraud* as a material accounting misstatement that violates GAAP and is made with the intent of misleading investors, and denote this latent construct by F^*. This definition is a property of a firm's reporting conduct rather than of any regulatory outcome. By contrast, the data provide only an enforcement label, L, indicating firms that the SEC both detects and formally identifies through an Accounting and Auditing Enforcement Release (AAER). Accordingly, L is an incomplete and selectively observed proxy for F^*. Our objective is therefore to recover signals that are informative about the underlying construct beyond the subset revealed through AAERs.

The mismatch between the latent construct F^* and the observed label L gives rise to two fundamental challenges for fraud detection: model training and model evaluation. A model trained directly on L may learn patterns associated with the SEC's enforcement rather than the underlying phenomenon F^*. On the flip side of the same coin, conventional out-of-sample classification accuracy cannot determine whether a model has recovered information about F^* or merely reproduced historical enforcement patterns represented by L. Nor is this problem unique to SEC enforcement. Alternative labels based on private litigation are likewise selectively generated because securities class actions disproportionately target large firms experiencing salient stock-price declines (Donelson et al., 2021). More broadly, these challenges arise whenever observed labels are generated through an endogenous institutional revelation process rather than by direct observation of the underlying economic state.

Because the training problem motivates the design of our method, we discuss it first before turning to the evaluation problem. We approach the problem by adopting a simple principle: **future proves past**. The central idea is that future accounting outcomes occupy a qualitatively—not merely quantitatively—different informational position from contemporaneous accounting variables because they emerge only after earlier reporting choices have interacted with economic reality. If current reports are manipulated, contemporaneous accounting numbers may already embody those reporting distortions. Future outcomes, by contrast, reflect what subsequently unfolds—operating performance, impairments, financing pressure, auditor scrutiny, and realized investment outcomes. Although noisy, these realizations may reveal tensions embedded in earlier financial reports that are not fully observable at the time they are issued. Because we apply this principle only within the training sample, however, model evaluation remains genuinely out-of-sample and free of look-ahead bias.

The intuition is fundamentally intertemporal. Fraud temporarily borrows from the future. Firms may inflate current performance through revenue inflation, expense understatement, or asset overstatement, but these accrual-based distortions cannot be sustained indefinitely and eventually reverse as economic reality catches up (Healy, 1985; Dechow et al., 1995). As the distortions unwind, subsequent accounting outcomes may exhibit characteristic patterns such as negative accruals, widening gaps between earnings and cash flow, deteriorating profitability, investment problems, and reductions in receivables or inventory (Dechow et al., 1996, 2011; Beneish, 1999; Kedia and Philippon, 2009).

Future outcomes, however, do not prove fraud. Many firms experience deteriorating performance or investment problems for ordinary economic reasons, while some fraudulent firms may avoid pronounced subsequent reversals. A procedure that mechanically relabels firms based on poor future performance would likely confuse fraud with financial distress, adverse shocks, or bad luck. Accordingly, our interpretation of the future-"proves"-past principle is deliberately modest. We use future outcomes only as an auxiliary source of information to make modest refinements to the supervisory signal during training, while AAER cases remain the anchor for confirmed enforcement actions. The objective is not to reclassify a new subset of firms as fraudulent ex ante, but rather to make the supervisory signal slightly more informative across the entire sample of firms. While each adjustment is modest, collectively they are intended to better align the observed training signal with F^* across the entire sample. The underlying intuition is that many machine learning methods learn from the structure of the entire training sample, so modest but broadly distributed improvements in alignment may be more valuable than large adjustments affecting only a small subset of firms.

While the training problem concerns how to recover a better supervisory signal from imperfect labels, the evaluation problem concerns how such models should be judged when those same labels are themselves incomplete. Simple assessment against L alone cannot establish whether a model has recovered information about F^* or merely reproduced historical enforcement patterns. We therefore evaluate the model using economically distinct validation tests outside the AAER label set. If the model successfully recovers information about F^*, firms predicted to be high risk should exhibit behaviors and outcomes independently associated with fraudulent reporting, even without formal enforcement actions. Thus, our framework combines complementary training and evaluation components: future accounting outcomes refine the supervisory signal during training, while economically distinct validation tests assess whether the recovered signal generalizes beyond observed enforcement actions.

Our sample period spans 1990–2019. We use windows ending by 1998 (with label information through 1999) as the initial training sample and reserve 2000–2019 for out-of-sample testing. A firm-year (i, t) receives a label of 1.0 if it is a confirmed AAER case. All remaining firm-years are initially assigned the neutral label of 0.5. We then refine these neutral labels for the *training* sample by incorporating accounting information from future years $\{t+1, t+2, t+3\}$, requiring that $t+3 \leq 1999$. Specifically, we apply a logistic transformation to future accounting outcomes after orthogonalizing them with respect to firm size, producing the *Soft Label*.[1] Confirmed AAER cases retain a label of exactly 1.0, whereas non-AAER firm-years receive adjusted labels centered around the neutral value of 0.5.

Fraud detection presents a challenging learning problem because fraud signals are weak, noisy, high-dimensional, and exhibit substantial temporal structure. To extract these signals, we use a residual convolutional neural network (CNN) that performs firm-year fraud detection using the previous five years of 23 accounting variables, treating each firm-year as a 5×23 accounting matrix that preserves temporal structure and captures nonlinear interactions across variables and over time. We further incorporate cost-

[1] This "look forward" procedure is applied only within the training sample.

sensitive resampling and dropout regularization to improve statistical signal recovery under the highly imbalanced class distribution characteristic of fraud detection.

Our main approach combines the CNN architecture with the soft-label framework described above, which we refer to as the *Soft Label* CNN. As a benchmark, we estimate the same CNN architecture using only the conventional AAER labels, referred to as the *Hard Label* CNN. We evaluate all models using standard out-of-sample fraud detection metrics. In addition to the Hard Label CNN, we compare our approach against RUSBoost (Bao et al., 2020), a state-of-the-art machine-learning benchmark for fraud detection, and the Dechow et al. (2011) F-score, the standard accounting-ratio fraud predictor, re-estimated fully out of sample under the same expanding-window protocol. The Hard Label CNN provides a strong baseline and substantially improves detection performance relative to existing approaches in the literature, increasing detection rates by a factor of two or more. The Soft Label CNN, however, shifts the learning objective from reproducing observed SEC enforcement labels toward identifying latent fraud under selective enforcement. The validation tests that follow are designed to evaluate whether this shift in the learning objective improves the identification of latent fraud.

The first two validation tests use externally screened, relatively clean populations as negative benchmarks. Arthur Andersen client firm-years from 2002–2006 that emerged from the intense post-Enron forensic scrutiny without enforcement actions provide a post-scrutiny benchmark of accounting quality. Bank-loan holders identified through DealScan provide a second benchmark because they have passed creditor screening and remain subject to ongoing covenant monitoring.

Consistent with its objective of identifying latent fraud rather than reproducing enforcement outcomes, the Soft Label CNN systematically avoids flagging both groups as high risk. Within the large-firm (> \$750M) sample, post-scrutiny Arthur Andersen clients are 2.4 percentage points less likely to be flagged than other large firms (logit $z = -2.12$; randomization inference $p = 0.017$), making the Soft Label CNN the only one of the three machine-learning models to exhibit a statistically significant tilt at conventional levels. Similarly, only 5.4% of bank-loan holders are flagged, compared with

15.7% of non-borrowers. All enforcement-trained alternatives reverse this raw pattern, flagging bank-loan holders more frequently than non-borrowers; for the Hard Label and RUSBoost that reversal reflects the concentration of borrowers among large firms and weakens or disappears once size is held fixed, so the unconditional avoidance is what is specific to the Soft Label. The third validation test examines seasoned equity issuance. Firms with temporarily overstated performance may have incentives to issue equity before the overstatement reverses (Loughran and Ritter, 1995; Teoh et al., 1998; Baker and Wurgler, 2002). Soft-label top-decile firms complete seasoned equity offerings at three times the base rate (27.1% over the next three years versus 9.0% for the bottom 90%, an 18.1 percentage point difference) and subsequently experience large negative post-issuance abnormal returns (−11.9% annualized FF25-adjusted returns over the $[t+1, t+5]$ window). In contrast, the Hard Label and RUSBoost show little association with equity issuance, and although the size-neutral F-score does flag future issuers, none of the three exhibits comparable post-issuance underperformance.

The fourth validation test examines whether predicted fraud probabilities identify future stock underperformance. A long–short portfolio that shorts the top decile of predicted fraud probability and holds the bottom 90% earns economically and statistically significant factor-adjusted alpha under the Soft Label CNN (8.4% under the Fama–French five-factor model, significant at the 1% level), whereas comparable portfolios formed using the enforcement-trained alternatives do not.

The four validation exercises rely on conceptually distinct sources of evidence—firm behavior, audit outcomes, creditor screening, and stock returns—yet all point in the same direction. Additional evidence comes from investor sentiment. The Soft Label CNN is the only one of the three models in the industry panel whose flag rate rises with the Baker and Wurgler (2006) investor-sentiment index precisely in the hard-to-value industries where sentiment-driven fraud should surface (Wang et al., 2010). Our results remain robust across alternative soft-label constructions, ranking rules, scaling choices, and training configurations (Appendix B). Taken together, the evidence suggests that the future-proves-past principle provides a more informative learning signal than observed enforcement labels alone, enabling machine learning models to better identify the underlying latent fraud construct, F^*.

A key feature of our design is that all validation tests are conducted entirely within the non-AAER sample ($F^* \setminus L$), allowing us to assess whether the model recovers economically meaningful information beyond the observed enforcement set. Our contribution is therefore not simply a more accurate fraud detector. Existing methods are designed to recover observed enforcement outcomes, and evaluated on that objective they perform well. Our results instead argue for a change in perspective: once enforcement labels are recognized as an incomplete proxy for the underlying fraud process, the learning objective itself becomes an object of study. The soft-label framework developed here represents one step in that direction, but more broadly this perspective motivates new approaches to learning and evaluating latent fraud.

The paper contributes to four strands of literature. It builds on the literature on hidden fraud and selective enforcement (Dyck et al., 2024) by treating the wedge $F^* \setminus L$ as the learning target. It extends the accounting literature on fraud and misstatement detection (Beneish, 1999; Dechow et al., 1995, 2011) and the growing machine-learning literature in accounting (Larcker and Zakolyukina, 2012; Brown et al., 2020; Bao et al., 2020) by introducing a deep learning framework that exploits the temporal structure of multi-year accounting data. Finally, it connects to the literature on the economic consequences of fraud and misreporting (Kedia and Philippon, 2009; Feroz et al., 1991; Loughran and Ritter, 1995; Teoh et al., 1998), since our validation tests examine whether model-identified firms exhibit the characteristics of hidden fraud firms before, or in the absence of, formal enforcement actions, and whether they subsequently exhibit the economic consequences of fraud and misreporting.

The remainder of this paper is organized as follows. Section 2 describes the data and sample construction. Section 3 develops the soft-label training target, the CNN implementation, and the training procedure, and introduces the benchmark models. Section 4 presents the out-of-sample detection results and documents the size bias of enforcement-trained models. Section 5 presents the four-test validation battery: the Arthur Andersen and bank-loan screening tests, equity issuance and post-issuance returns, and long-short portfolio alphas. Section 6 ablates the training target into its components and examines the investor-sentiment predictions. Section 7 concludes.

2 Data and Sample Construction

2.1 Defining Fraud

Corporate fraud in the broad sense spans a wide range of misconduct: insider trading, Ponzi schemes, benchmark manipulation, and the opportunistic timing of option grants, among others (Lie, 2005; Heron and Lie, 2007; Lie, 2025). Our object of interest is narrower: fraudulent *financial reporting*. Following Dechow et al. (2011), we define *fraud* as a material, economically significant accounting misstatement that (i) violates GAAP and (ii) is made with the intent of misleading investors. We refer to this latent construct as *latent fraud*. Fraud must be defined independently of SEC enforcement; otherwise, undiscovered fraud would be impossible by definition. Latent fraud is therefore not directly observable: some fraudulent misstatements are never detected, and some detected misstatements are never formally designated.[2]

Our empirical measure is the *enforcement label*: the part of latent fraud that the SEC both detects and designates through an Accounting and Auditing Enforcement Release (AAER). By construction, an AAER label is the *joint* outcome of three events—a misstatement occurs, the SEC detects it, and the SEC designates it in an enforcement release—so a firm-year carries the label only if all three hold (Dechow et al., 2011). The intent element of the definition enters the data only at the second and third stages: what we observe is the SEC's allegation that the misstatement was intentional, not intent itself, so enforcement is the measurement apparatus for latent fraud rather than part of

[2]Following Dechow et al. (2011) and Dyck et al. (2024), we use the term *fraud* as shorthand for an intentional material misstatement. For the observed enforcement label defined below, the intent element is measured by the SEC's allegation, in the enforcement release, that the misstatement was intentional; consistent with the enforcement setting, the firms and managers involved typically neither admit nor deny guilt, and we make no claim of adjudicated guilt for any individual firm. Dechow et al. (2011) use the term *misstatement* for the same reason.

its definition. The advantage of the enforcement label is its reliability: a firm-year that carries it is highly likely to be a genuine misstatement. Its limitation is censoring: Dyck et al. (2024) estimate that in normal times only about one-third of corporate fraud is ever detected, so the enforcement label is a selected, incomplete proxy for latent fraud.[3]

Fraud that occurred but escaped enforcement—the wedge between latent fraud and the enforcement label—is precisely the hidden fraud this paper sets out to detect (Figure 1 depicts this structure). The relation between the two is one of strict containment: every labeled firm-year is genuine fraud, but genuine fraud extends beyond the labeled firm-years. The training and evaluation problems of Section 1 are the two consequences of that containment being strict: training on the enforcement label risks learning the enforcement process rather than the fraud process, and accuracy measured against the enforcement label cannot certify detection of the fraud it leaves out.

One definitional caveat deserves emphasis at the outset. Latent fraud as defined above is narrow: it requires both a GAAP violation and intent, and neither element is observable outside the enforcement process. The forward-looking indicators underlying the soft label (Section 3.1) therefore cannot target latent fraud exactly; what they capture is a broader latent construct, aggressive accounting that subsequently reverses, which contains the hidden fraud we seek but may also contain within-GAAP earnings management that involves neither a violation nor demonstrable intent to mislead. This gap is not specific to our construction: because intent is unobservable, *any* accounting-based signal for hidden fraud is necessarily a signal for a superset of it. What can be assessed is whether the firms the signal ranks highest behave like fraud firms rather than like merely aggressive or distressed ones, and this is precisely the role of the validation exercises in Sections 5–6: opportunistic equity issuance ahead of reversals, avoidance of externally screened (relatively clean) populations, and negative risk-adjusted returns are patterns associated with misleading reporting rather than with aggressive-but-informative reporting or ordinary deterioration.

[3]The incompleteness is measurable within the *detected* part of latent fraud itself: comparing public and private enforcement over 1998–2014, Donelson et al. (2021) find that roughly two-thirds (64%) of settled securities class actions alleging accounting fraud have no SEC enforcement counterpart. Even among frauds credibly detected by *some* channel, the AAER label designates only a minority.

The remainder of this section describes how we operationalize the enforcement label from the AAER record; Section 3 then describes how the soft label uses forward-looking accounting outcomes to recover information about the fraud that lies beyond it.

2.2 Data Source

Our fraud sample is constructed from the Accounting and Auditing Enforcement Releases (AAERs) database compiled by Dechow et al. (2011). Since 1982, the SEC has issued AAERs during or at the conclusion of investigations against companies, auditors, or officers for alleged accounting and/or auditing misconduct. We use the version updated through December 31, 2021 (AAER No. 1 through No. 4,278), which covers 1,816 unique firm misstatement events. Among these, 1,087 events involve financial misstatements that affect at least one quarterly or annual financial statement. The annual file ("ann" sheet) records each fiscal year in which a misstatement occurred as a separate observation, yielding 2,195 firm-year observations across 942 distinct firms.

2.3 Sample Selection

Following Dechow et al. (2011), we apply the following filters to the annual file:

1. **Overstatement only.** We exclude firm-years in which earnings or revenues were understated, retaining only overstatement cases. This removes 56 firm-year observations (11 firms), leaving 2,139 firm-years (931 firms).

2. **Compustat identifier (GVKEY).** We require each firm to have a valid GVKEY for matching to Compustat. This removes 384 firm-years (187 firms), leaving 1,755 firm-years (744 firms).

3. **Compustat match with non-missing total assets.** We merge the remaining sample with Compustat annual data by GVKEY and fiscal year, and require total assets (*at*) to be non-missing. This removes 136 firm-years (42 firms), leaving 1,619 firm-years (702 firms).

4. **Exclude financial firms.** We exclude banks, insurance companies, and other financial firms (SIC codes 6000–6999), as their financial statement structure differs substantially from non-financial firms and many accrual-related variables are not applicable. This removes 212 firm-years (93 firms), leaving 1,407 firm-years (609 firms).

After applying these four filters, we obtain 1,407 fraud firm-year observations from 609 distinct firms, covering fiscal years 1971 through 2019.

2.4 Sample Construction for Prediction

Our prediction model requires each fraud event to have five consecutive years of complete financial data ending in the misstatement year. We apply the following additional filters:

5. **Collapse to unique GVKEY firm-years.** The annual file records each fiscal year of each AAER as a separate observation, so nine records are duplicates at the firm-year level (two distinct AAERs covering the same fiscal year of the same firm), and several company records map to the same Compustat GVKEY. Collapsing to unique GVKEY firm-years leaves 1,398 fraud firm-years from 597 distinct GVKEYs.

6. **Analysis window restriction.** We restrict the sample to fiscal years 1990 through 2019, the analysis window used throughout the paper (Compustat coverage in earlier years is sparse, and the five-year rolling window requirement effectively bars pre-1985 events anyway). This removes 205 fraud firm-years (105 firms), leaving 1,193 fraud firm-years (492 firms).

7. **Five-year window requirement.** For each fraud event, we require at least five years of Compustat data up to and including the misstatement year. This removes 181 fraud events (85 firms) that lack sufficient history, leaving 1,012 events.

8. **Non-missing key variables.** We require 23 financial statement items to be non-missing across all five years of the window. The most frequently missing items are

sale of stock ($sstk$), current assets (act), current liabilities (lct), and income taxes payable (txp). This removes 229 fraud events (102 firms).

9. **Consecutive fiscal years.** We require the five years in each window to be consecutive (i.e., no gaps in fiscal year). Once duplicate records are collapsed, this requirement drops no additional events.

2.5 Final Sample

After applying the Section 2.3 filters and the five-year window requirement, the AAER side of the panel contains **783 AAER fraud firm-years from 305 distinct firms** in the 1990–2019 analysis window. Each observation (i, t) uses the input window $\{t-4, \ldots, t\}$ (the five consecutive years of accounting data fed to the CNN) and is paired with a label window $\{t+1, t+2, t+3\}$ used to construct the soft label (Section 3.1). Note that the input window and the soft-label window are separated by only one year (input ends at t, label starts at $t+1$); we describe the additional safeguard that prevents this short label-window gap from leaking test-year accounting data into training in Section 3.4.3. The non-fraud side of the panel requires soft-label coverage: the label construction (Section 3.1) requires a Compustat–CRSP match carrying the market and book equity it needs to place the firm in a size–book-to-market portfolio and to form its market-based residualization controls, and a five-year window enters the panel when any of its years carries a label (the most recent labeled year is used). Financial firms are in effect excluded by the 23-variable input requirement itself (banks and insurers lack several of the required working-capital items), so the resulting panel is more than 99% non-financial and non-utility at the terminal year. The full panel used for training comprises 84,089 firm-years (783 AAER fraud, 83,306 non-fraud); the models score 57,635 firm-years over 2000–2019 (Table 3), of which the 54,070 outside financials and utilities form the evaluation pool (493 AAER firm-years; see Table 4). Table 3 reports the annual distribution; fraud rates peak in the early 2000s coinciding with the Enron/WorldCom era and decline steadily thereafter.

Our dataset exhibits severe class imbalance, a fundamental characteristic of fraud detection research. As shown in Table 3, fraud firm-years peak at 2.2% of the analysis sample (in 2001) and never exceed that share in any other year, and the pooled fraud rate is 0.93%. This extreme imbalance motivates the soft-label design that provides a continuous signal across non-AAER firm-years (Section 3.1) and the data-augmentation strategy described in the training procedure (Section 3.4).

3 The Soft Label and Model Design

3.1 Soft Label Construction

The standard approach in fraud detection trains models using binary labels: a firm-year is labeled 1 if it is subject to an SEC Accounting and Auditing Enforcement Release (AAER) and 0 otherwise, the *hard label* method. Section 2.1 established why this target is contaminated: Dyck et al. (2024) estimate that at least 10% of large publicly traded firms engage in fraud, while AAER records cover only about 1% of firms in our sample, so the "0" class mixes genuinely clean firms with undetected fraudulent ones, and the binary label treats them identically.

We propose a *soft label* method that operationalizes the future-proves-past principle of the Introduction: it preserves the AAER label on known AAER firms while replacing the uniform "0" target for non-AAER firms with a continuous measure of forward-looking accounting anomaly. Rather than using future stock returns, which conflate fraud-related information with a wide range of non-accounting shocks, we construct the soft label from forward-looking *accounting* variables that directly capture the correction patterns (accrual reversals, profitability deterioration, and write-downs) through which manipulation unwinds.

In the language of modern machine learning and econometrics, this construction simultaneously addresses three interrelated identification problems. **First, noisy-label correction**: the binary AAER label is contaminated in one direction (hidden fraud mislabeled as zero), and the soft label functions as an auxiliary supervisory signal that partially de-noises the non-AAER class (Natarajan et al., 2013). **Second, weak supervision**: the four forward-looking indicators are not exact fraud labels but *noisy* *proxies* of the latent fraud state, in the spirit of distantly supervised and Snorkel-style labeling functions (Ratner et al., 2017). Each indicator individually is a weak signal of misreporting; their aggregation into an integer count c of binary flags converts four weak signals into a more reliable composite. **Third, latent-state inference under selective enforcement**: SEC enforcement is itself selective and size-biased, so the observed AAER label is a non-random sample of the latent fraud population (Dyck et al., 2024). The forward indicators are generated outside the SEC's selection process and thus partially purge the selection bias from the supervisory signal. The four-step construction below operationalizes these ideas jointly.

Step 1. Four forward-looking accounting indicators. For each firm-year (i, t), we construct four indicators $Y_{1,i,t}$ through $Y_{4,i,t}$ that capture forward-looking accounting reversal and deterioration patterns. We adopt a uniform sign convention: *lower values indicate worse (more fraud-like) outcomes*. The four indicators are:

- Y_1 *cash-flow-statement accruals*: $Y_{1,i,t} = -\text{Accruals}_{i,t}$, where $\text{Accruals}_{i,t} = (ib_t - oancf_t)/at_{t-1}$ is the gap between net income and operating cash flow scaled by lagged total assets. Large positive accruals (earnings well above cash flow) signal earnings management; the sign flip ensures large accruals map to low Y_1.

- Y_2 *accrual reversal*: $Y_{2,i,t} = -(\text{Accruals}_{i,t} - \text{Accruals}_{i,t-1})$, the year-on-year change in $\text{Accruals}_{i,t}$ defined above. A growing accruals gap maps to low Y_2.

- Y_3 *ROA deterioration*: $Y_{3,i,t} = \text{ROA}_{i,t} - \text{ROA}_{i,t-1}$, with $\text{ROA}_{i,t} = ib_t/at_{t-1}$. ROA falling year-on-year produces a negative Y_3, consistent with the convention.

- Y_4 *unexplained investment*: $Y_{4,i,t} = -|\widehat{\varepsilon}_{i,t}|$, where $\widehat{\varepsilon}_{i,t}$ is the residual from the within-(year, FF48 industry) cross-sectional OLS

$$\frac{capx_{i,t}}{at_{i,t-1}} = \alpha_{j,t} + \beta_{1,j,t}\, Q_{i,t} + \beta_{2,j,t}\, CF_{i,t} + \beta_{3,j,t}\, Size_{i,t} + \varepsilon_{i,t}, \qquad (1)$$

with Tobin's Q $Q_{i,t} = (|prcc\_f_{i,t}| \cdot csho_{i,t} + lt_{i,t})/at_{i,t}$, cash flow $CF_{i,t} = oancf_{i,t}/at_{i,t-1}$, and size $Size_{i,t} = \log(at_{i,t})$; j indexes Fama–French 48 industries (financials and utilities excluded), and each (year, industry) cell with at least 20 non-missing observations is fit separately. Investment decisions poorly explained by Q, cash flow, and size produce large $|\widehat{\varepsilon}_{i,t}|$ and hence low Y_4.

Each $Y_{k,i,t}$ is winsorized at the 1st and 99th percentiles. Each indicator is then *residualized*: for every fiscal year we regress $Y_{k,i,t}$ cross-sectionally on a control vector containing the Altman Z-score, return on assets, leverage, log total assets, market-to-book, cash holdings, the trailing six-month stock return, and the trailing twelve-month return volatility, plus Fama–French 48 industry indicators, and replace $Y_{k,i,t}$ with the regression residual (the lower-is-worse orientation is preserved; we retain the notation $Y_{k,i,t}$ for the residualized values in the steps below). This regression stage purges the components of the raw indicators that are predictable from size, distress, valuation, liquidity, and recent stock-price behavior, so that the forward window aggregates *abnormal* reversals rather than mechanical correlates of firm condition; the within-(year, size-decile) ranking of Step 3 then adds a non-parametric size guarantee on top.

Step 2. Forward three-year aggregation. For each base year t and each indicator $k \in \{1, \ldots, 4\}$, we aggregate $Y_{k,i,s}$ across the strictly forward three-year window $\mathcal{T}_{i,t} \subseteq \{t + 1, t + 2, t + 3\}$, requiring non-missing values in at least two of the three years. Indicators capturing accruals magnitude or year-on-year accounting reversals (Y_1, Y_2, Y_3) are aggregated by the within-window *minimum*: earnings-management episodes typically appear as a single charge-laden year, so the worst observed year is most informative, and a within-window mean would let a positive year cancel a severely negative one. The unexplained-investment residual (Y_4) is aggregated by the within-window *mean*: investment deviations from fundamentals tend to unwind gradually over multiple years, so sustained levels are more informative than single-year spikes. Formally,

$$\bar{Y}_{k,i,t} = \begin{cases} \min_{s \in \mathcal{T}_{i,t}} Y_{k,i,s}, & k \in \{1, 2, 3\}, \\ \frac{1}{|\mathcal{T}_{i,t}|} \sum_{s \in \mathcal{T}_{i,t}} Y_{k,i,s}, & k = 4. \end{cases} \tag{2}$$

Step 3. Size-decile-conditional ranking and binary flagging. A direct cross-sectional ranking of $\bar{Y}_{k,i,t}$ would mechanically tilt the soft label by firm size if any of the four aggregated indicators carries a size-related drift. To produce a label that is size-orthogonal *by construction*, we form size deciles within each fiscal year on $\log(at_{i,t})$ at the base year, and then rank each indicator within each (year, size-decile) cell:

1. For each base year t, partition firms into ten equal-sized bins by $\log(at_{i,t})$, indexed by $d(i,t) \in \{1,\ldots,10\}$.

2. Within each (year, size-decile) cell with at least 30 observations, compute the within-cell empirical-CDF rank of $\bar{Y}_{k,i,t}$ for each k separately, with rank ascending in $\bar{Y}_k$ (so the lowest 10% by rank are also the 10% lowest values of $\bar{Y}_k$, which by the Step 1 sign convention are the most fraud-like on indicator k).

3. Define $F_{k,i,t} = \mathbf{1}\{\text{within-cell rank of } \bar{Y}_{k,i,t} \leq 0.10\}$, equal to one when firm i falls in the worst-10% sub-cell on indicator k relative to its size-decile peers in base year t, and zero otherwise.

By construction, the flag rate is exactly 10% per indicator within every (year, size-decile) cell, so any size-tilt in the underlying $\bar{Y}_k$ values is absorbed by the cell stratification rather than propagating to F_k. We then aggregate the four flags into a single integer count:

$$c_{i,t} \;=\; \sum_{k=1}^{4} F_{k,i,t} \;\in\; \{0,1,2,3,4\}, \tag{3}$$

where firm-years with any missing flag are dropped to ensure a clean count interpretation.

Step 4. Logistic transformation and AAER anchor. We map the count $c_{i,t}$ to a soft label via a centered logistic transform with scaling parameter $\beta = 0.05$, centered at the count midpoint $c = 2$ so that $c_{i,t} = 2$ yields exactly 0.5:

$$\text{Soft Label}_{i,t} \;=\; 1 \;-\; \frac{1}{1 + \exp\left(-(2 - c_{i,t}) \cdot \beta\right)}. \tag{4}$$

Concretely, Soft Label $\in \{0.475,\ 0.488,\ 0.500,\ 0.512,\ 0.525\}$ for $c \in \{0,1,2,3,4\}$. AAER firm-years (those with $\text{AAER}_{i,t} = 1$ in the training data) are assigned a label of exactly 1.0, overriding the logistic value. Non-AAER firm-years receive the count-based logistic value, which lies in a narrow band $[0.475, 0.525]$ around the neutral 0.5.

Two design choices deserve note. First, the binarization in Step 3 accepts some information loss, relative to a continuous within-cell z-score composite, in exchange for a clean size-orthogonality guarantee: each F_k has a flag rate of exactly 10% within every (year, size-decile) cell, so the count c cannot mechanically inherit any size tilt from the underlying $\bar{Y}_k$ distributions. Second, the small β keeps the soft label a gentle modifier of the neutral 0.5 target rather than a dominant signal that could overwhelm the AAER anchor at 1.0. We use the soft label as the training target; the hard label (binary AAER indicator) serves as the within-architecture benchmark throughout our analysis.

Methodological Position

Because hidden fraud is fundamentally unobservable, the contribution of the paper cannot rest primarily on classification accuracy or on the exact specification of the soft-label construction. In a latent-state environment with contaminated labels there is no verifiable record of actual fraud against which alternative constructions can be definitively compared; recovering latent accounting manipulation must therefore rely on imperfect proxy signals and weak supervision rather than exact classification. The informational role of the soft label derives not merely from injecting additional accounting variables but from the fact that the auxiliary signal is *forward-looking* rather than contemporaneous. This distinction is qualitative, not merely quantitative, and is the precise sense in which future proves past. Contemporaneous accounting variables are generated within the same reporting regime as any potential manipulation; if a firm is misreporting, those numbers *are* the misreported numbers. Future accounting outcomes occupy a fundamentally different informational position, because intertemporal accounting consistency and underlying economic constraints eventually exert pressure on the reporting process: revenue inflation, expense understatement, overstated asset values, and overly optimistic investment cannot be sustained indefinitely without future correction, reversal, or operational deterioration. Future realizations therefore act as partial *reality constraints* on the latent accounting state behind earlier reports.

The empirical results below are not driven by a narrow specification choice. We conduct sensitivity analyses varying the indicator composition (substituting alternative correction-pattern indicators), the aggregation rule (MIN vs. MEAN), the within-cell ranking threshold (top 5%, 10%, 20%), the logistic scaling parameter β, the size-decile granularity, and the training configuration (architecture variants, fraud-replication multiplier, and certainty-weight bounds; the full ten-configuration hyperparameter grid and the pre-committed validation rule that selects the reported configuration are documented in Appendix B). The core qualitative findings remain robust across these alternative constructions; the contribution does not rest on a finely tuned specification but on the broader idea that future accounting reversals contain economically meaningful information about latent accounting problems that standard hard-label training procedures leave unexploited.

Figure 2 summarizes the resulting information timeline: for each training observation, future accounting information enters the training *label* only, never the model inputs, and the leakage-control rules described in Section 3.4.3 guarantee that no information dated at or after the test year influences any training label.

The central empirical burden of the paper therefore falls on the validation exercises of Sections 5–6, which assess whether the resulting signal behaves like hidden fraud rather than generic deterioration or distress.

3.2 Dechow F-score Benchmark (Out-of-Sample Re-estimation)

Alongside the Hard Label CNN and RUSBoost (Bao et al., 2020), we include the Dechow et al. (2011) F-score (their Model 1) as a fourth benchmark. It is the standard accounting-ratio fraud predictor in the literature. Because it is a parsimonious logit on seven financial-statement ratios and uses no firm-size variable, it provides a transparent, size-neutral comparison for the machine-learning models. To place it on exactly the same footing as the Soft Label, Hard Label, and RUSBoost models, we do *not* import Dechow et al.'s published coefficients; instead we *re-estimate* the model out of sample, under the identical expanding-window, retrain-per-year protocol used for the neural networks, on the identical firm-year sample and with the same binary AAER label.

Model 1 uses seven financial-statement variables (RSST accruals, the change in receivables, the change in inventory, the percentage of soft assets, the change in cash sales, the change in return on assets, and an actual-issuance indicator), each constructed from Compustat exactly as specified in Dechow et al. (2011), Table 3. For each test year $\tau \in \{2000, \ldots, 2019\}$ we estimate the logit by maximum likelihood on the expanding window of all real (non-replicated) firm-years with $1990 \leq t \leq \tau - 2$; the fitted coefficients are then applied to the held-out firm-years with $t = \tau$ to produce out-of-sample predicted probabilities. The six continuous variables are winsorized at their 1st and 99th percentiles, with the cut-points estimated on the training sample only. In all cross-model comparisons we rank firms within each year on the predicted probability (equivalently on the F-score, since the two are monotone within a year) and restrict the output to the identical test universe used by the other models; because Model 1's inputs are occasionally missing, the F-score is defined for 53,743 of the 54,070 test firm-years (99.4%) and is ranked within year on the non-missing set.

Before deploying the F-score as an out-of-sample benchmark, we verify that our variable construction reproduces Dechow et al. (2011): estimating their three nested logit models in sample on the strict-paper universe (overstatement AAERs with case number $\leq$ 2261; all Compustat firm-years 1979–2002) recovers coefficients that match the published Table 7 values in sign and magnitude across all three models and reproduces the paper's headline diagnostics, including an Enron (fiscal 2000) Model-1 F-score of 2.65 against the paper's 2.76. This confirms that the Model-1 inputs we feed into the out-of-sample re-estimation are built correctly.

3.3 Model Architecture

The choice of a Convolutional Neural Network is motivated by the requirements of the latent-fraud detection problem rather than by a general preference for deep learning. Our central empirical challenge is not simply class imbalance but *label contamination*: the observed AAER label is a selectively observed enforcement outcome, while many non-AAER firms may contain hidden fraud. Addressing this problem requires a framework that can ingest soft, probabilistic, or weak-supervision targets for non-AAER firms; standard hard-label classifiers, including RUSBoost (Bao et al., 2020), are designed around binary outcomes rather than the graded targets that are central to our approach.

A second consideration is that detecting hidden fraud is a low-signal, high-noise prediction problem. The relevant information is unlikely to reside in any single accounting ratio or isolated year; latent manipulation more plausibly appears through nonlinear interactions among accounting variables and through their evolution over multiple years.

The CNN provides such a framework. We represent each firm-year as a 5×23 matrix (five consecutive years $\times$ 23 Compustat variables; Table 1, Panels A–D), so that the network can learn local patterns across variables and over time without requiring all interactions to be specified ex ante (Figure 3 sketches the architecture). This flexibility is especially important in our setting because the goal is not merely to reproduce the observed SEC enforcement decisions, but to recover latent accounting-risk patterns that enforcement may miss. The CNN therefore serves as an implementation device that allows the soft-label idea to operate in a high-dimensional, nonlinear, and temporally structured environment. The architecture combines two convolutional blocks with a residual connection (LeCun et al., 2015; He et al., 2016) and a focal-loss objective adapted to the mixed hard/soft label structure; Figure 3 illustrates the complete architecture, and Appendix A reports the implementation details and the certainty-weighted focal-loss specification.

A deliberate feature of the architecture is the asymmetric treatment of the variable and time dimensions. The convolutional kernels span adjacent *variables* within a single fiscal year (kernels of height one in the year dimension), with weights shared across the five years, while all *cross-year* linkages are formed by the position-specific fully connected layers that follow. For this adjacency to be economically meaningful, the 23 variables are arranged in financial-statement order (assets, then liabilities and equity, then income-statement items, then cash-flow and market items; Table 1, Panels A–D), each block in standard accounting sequence, so that columns adjacent in the input matrix are related items: cash beside receivables beside inventory, or sales beside cost of goods sold. The within-year kernel therefore spans coherent local groups of accounts rather than an arbitrary column ordering. The economic logic is that the two dimensions carry different kinds of structure. Within a fiscal year, financial-statement variables are bound by accounting identities and ratio relationships (accruals against operating cash flow, receivables against sales) that mean the same thing in every year of the window; sharing the within-year filters across years imposes exactly this stationarity and conserves parameters. Across years, by contrast, the input window is anchored at the prediction year: a misstatement sits at the end of the window, so the same accounting dynamic carries different information in year $t-1$ than in year $t-4$ as borrowed prosperity builds toward the terminal year. Position-specific dense weights preserve this calendar-position information. This is also why we do not convolve over the time dimension. A temporal convolution would impose time-translation invariance, forcing an accrual build-up four years before the prediction date to receive the same weight as one immediately preceding it, and thereby sacrificing exactly the calendar-position information that matters. With only five time steps, moreover, such a convolution is a constrained special case of the dense layer we already use, so it would add no expressive power in return. Keeping the temporal stage in the dense layers, with low capacity and heavy regularization, also limits overfitting risk given only 783 positive training events.

At the same time, the paper's contribution does not rest on the CNN architecture alone: the architecture is valuable because it enables the broader weak-supervision design, and the same design is in principle compatible with alternative function approximators suitable for tabular time-series inputs (e.g., transformer-style attention models, recurrent networks, or multi-layer perceptrons).

3.4 Training Procedure

3.4.1 Data Augmentation

To address class imbalance, we replicate every confirmed-fraud (AAER = 1) training window twenty times, so each fraud case enters training as twenty-one copies. Dropout regularization (Srivastava et al., 2014) ensures that each replicated copy is seen through a different effective subnetwork, so the replication functions as data augmentation rather than mere duplication of identical inputs.

3.4.2 Loss Function

During training, we combine observations with varying label certainty: hard labels ($y_i = 1$) for confirmed AAER fraud cases and soft labels ($y_i \in [0.475, 0.525]$) for non-AAER firm-years. To learn from this heterogeneous label structure, we use a Smooth Certainty-Weighted Focal Loss (Lin et al., 2017) that down-weights observations whose labels are uncertain (close to 0.5) and up-weights confident labels (close to 0 or 1):

$$\mathcal{L}_i = w(y_i) \cdot \alpha\,(1 - p_t)^{\gamma} \cdot \mathrm{BCE}(y_i, \hat{y}_i), \qquad w(y_i) = w_{\min} + (w_{\max} - w_{\min}) \cdot 2|y_i - 0.5|, \tag{5}$$

where $\hat{y}_i$ is the predicted probability, BCE is the binary cross-entropy loss, and $p_t = \exp(-\mathrm{BCE})$ is the predicted probability of the true class. The certainty weight $w(y_i)$ rises from $w_{\min}$ at $y_i = 0.5$ (maximum uncertainty) to $w_{\max}$ at $y_i \in \{0, 1\}$ (hard labels). Under our soft-label design, confirmed AAER cases receive the maximum weight, while non-AAER firm-years receive weights that increase monotonically with the magnitude of their forward-looking accounting-anomaly signal. Optimization uses the Adam optimizer with mini-batch gradient descent.

3.4.3 Out-of-Sample Scheme and Soft-Label Leakage Control

We use an expanding-window training scheme: for each test year $T \in \{2000, \ldots, 2019\}$, the model is trained on observations ending in $\{1990, \ldots, T-2\}$, with a two-year (24-

month) gap between training and test following Bao et al. (2020); we feed the model a five-year input window of the prior accounting data, and evaluate on observations whose input window ends at T. The out-of-sample evaluation pool contains 54,070 firm-years and 493 AAER firm-years (Table 4). For each test year, we follow Bao et al. (2020) and classify the top 1% of firms by predicted probability as flagged. The 1% cutoff mirrors their rationale: because typically fewer than 1% of firms in a year are AAER fraud cases, flagging the top 1% by predicted probability targets a fraction comparable to the base rate of enforced fraud, rather than relying on a classifier-specific probability threshold that would depend on the model and the training sample.

The two-year input-side gap is necessary but not sufficient when the training target is the soft label, because the soft label of a training observation with input window ending at t is computed from accounting variables in $\{t+1, t+2, t+3\}$ (Section 3.1, Step 2). For training samples with $t \in \{T-2, T-3\}$, this label window overlaps the test year T (e.g., $t = T-3$ yields label window $\{T-2, T-1, T\}$), so the unmasked soft label would carry information about year-T accounting outcomes that the model would not have at prediction time. We close this gap by replacing the soft label of any non-AAER training sample with $t \in \{T-2, T-3\}$ by the neutral value 0.5 during the training of test year T, removing all directional signal from those observations. AAER firm-years are unaffected because their label is the binary AAER indicator and does not depend on future accounting data. Training samples with $t \leq T-4$ have label window $\subseteq \{T-3, T-2, T-1\}$ and are used with their actual soft label.

4 Fraud Detection Performance

We organize the out-of-sample evidence into two distinct sets that play different roles in the paper. The first, the *detection* results of this section (Table 4 and Figure 4), compares model rankings directly against the SEC labels and measures each model's ability to recover known AAER cases. These tests serve as a necessary check that the soft-label target does not sacrifice SEC-labeled fraud capture. The second, the *validation* results of Sections 5 and 6 (Tables 5–11), compares model rankings against external signals that lie outside the SEC label set entirely: post-scrutiny Arthur Andersen status, bank-loan screening, future equity issuance behavior, post-issuance returns, long-short portfolio alpha, and investor-sentiment exposure. Because no data source records which firms committed the fraud that enforcement missed, the validation set carries the central identification burden of the paper: detection establishes that the soft label preserves SEC-labeled fraud capture, while validation establishes that it captures the hidden fraud the SEC misses. Throughout, four methods are ranked on the same test universe: our Soft Label CNN, the identically specified Hard Label CNN, RUSBoost (Bao et al., 2020), and the out-of-sample re-estimated Dechow et al. (2011) F-score (Section 3.2). The F-score enters the detection results and the descriptive comparisons; the regressions and the factor-model tests are estimated for the three machine-learning models.

4.1 Sample Characteristics

Table 3 presents the annual distribution of unique five-year sample windows from 1990 to 2019. The Step 1 pool comprises 135,897 firm-years from Compustat with a complete, consecutive five-year window of the 23 accounting variables listed in Table 1; the soft-label-coverage requirement (which embeds the CRSP common-equity screen of the label construction) reduces this to the 84,089-firm-year analysis sample used for training. The 783 AAER misstatement years (already filtered to non-financial in Table 2) are retained throughout. Annual coverage at Step 1 ranges from 3,969 windows (2019) to 5,301 (1999); the unconditional AAER fraud rate peaks in 2001 during the Enron/WorldCom era and falls sharply after 2010. This extreme class imbalance motivates both the soft-label design described in Section 3.1 and the fraud-case replication strategy described in Section 3.4.

For our out-of-sample evaluation, we train on data through fiscal year $T-2$ and test on T for each test year $T \in \{2000, \ldots, 2019\}$. The models score every 2000–2019 window whose terminal year carries a Compustat–CRSP match with market and book equity, 57,635 firm-years (Table 3); excluding financial and utility firms leaves the evaluation pool, 54,070 firm-years and 493 AAER firm-years, is identical for all four models, and is the basis for the detection results in Table 4. The cross-sectional and portfolio analyses (Tables 5–11) draw on subsets of this aligned sample as detailed in each section.

4.2 Detection Results

Table 4 reports out-of-sample fraud detection results for 2000–2019 on the evaluation pool of 54,070 firm-years (493 AAER fraud firm-years). We compare the Soft Label CNN (our proposed method), the Hard Label CNN (the same architecture trained with binary AAER labels), RUSBoost (Bao et al., 2020), and the out-of-sample Dechow et al. (2011) F-score. For each test year we classify the top 1% of firms by predicted fraud probability as flagged. Each model's score is its within-year ranking averaged across the 30 training runs, the run-noise-robust ensemble aggregate used for all ranking-based results in the paper. The table reports both recall (the detection rate, TP/(TP+FN)) and precision (TP/flagged); because the within-year top-1% rule flags the same fraction of firms for every method, the two measures move together.

One scope caveat applies to every model compared here: all four see only accounting numbers, so the relevant benchmark is the accounting-detectable part of latent fraud. Enforcement cases triggered by non-accounting conduct—textual disclosure, governance breaches, market manipulation—or by manipulation too subtle to surface in the financial statements lie outside the reach of any accounting-based model, however flexible, and place an upper bound on the detection rates reported below.

On the full aligned sample (Panel B), the yearly-mean detection rate is 14.5% for the Soft Label CNN, 18.3% for the Hard Label CNN, 6.5% for RUSBoost, and 8.3% for the F-score; pooled detection rates are 17.0%, 18.5%, 8.7%, and 3.7% respectively. The two neural-network models are broadly comparable (Hard slightly ahead, consistent with the hard label being trained directly on the evaluation target), and both detect known AAER cases at 1.8 to 5.0 times the rate of RUSBoost and the F-score, depending on which of the two summary measures is used. Restricting the evaluation to large firms with total assets above $750 million (Panel A)—the symmetric Dyck et al. (2024) threshold—

sharpens this pattern: the Hard Label detects about 21% and the Soft Label about 16% of large-firm AAER cases per year while RUSBoost and the F-score fall below 4%, a first indication that the benchmarks' full-sample detection interacts strongly with firm size, a pattern we examine directly in the next subsection.

One caveat concerns the label itself: the detected counts thin out in the later sample years partly because SEC accounting enforcement itself contracted sharply after the mid-2000s (Donelson et al., 2021), and recent misstatement years are further truncated by the enforcement lag. The label, not the models, loses power late in the sample.

These figures answer the primary detection question: the soft-label target gives up little SEC-labeled fraud capture relative to its hard-label twin, and both neural models recover known AAER cases at substantially higher rates than the machine-learning and statistical benchmarks. But they leave open the more interesting empirical question of *which* firms each model flags. Because the two neural models differ only in their training target (soft vs. hard labels) and are applied to the same test sample, any systematic differences in the rankings they produce are informative about how the soft label changes model behavior. We investigate this in Section 4.3 and in the validation tests of Sections 5–6.

4.3 Size-Controlled Detection and the Size Bias of Enforcement-Trained Models

With clean labels, the area under the curve (AUC) has a simple ranking interpretation: it is the probability that a randomly chosen fraud firm-year is ranked above a randomly chosen non-fraud firm-year. Two features of our setting complicate that interpretation. First, the measure treats every non-AAER firm-year as fraud-free, so a model that successfully surfaces hidden fraud is penalized for doing so: each hidden-fraud firm-year it ranks highly is scored as a false alarm. Second, SEC enforcement is heavily tilted toward larger firms, so a model trained on hard AAER labels inherits that size tilt and ranks large firms higher than a size-orthogonalized model would.

Appendix Appendix C repeats Table 4 for the 2000–2008 and 2000–2014 sub-windows. Holding firm size roughly fixed separates the models sharply—Panel A against Panel B of Table 4—and the reason is not the one a size tilt first suggests. Ranking by size alone recovers almost no enforced fraud at a one-percent cutoff: 0.6% of cases, below the 1% a random ranking would take, because the very largest firms are not where enforcement concentrates. What a size tilt buys is position rather than prediction. Firms above \$750 million are 39% of the sample but 53% of the enforced cases—a 1.25% enforced-fraud rate against 0.70% among smaller firms—and RUSBoost, whose scores correlate with size at $\rho = 0.49$ within year against the Soft Label's -0.12, places two-thirds of its top-decile flags among them. Restrict the pool to large firms and that position is gone: its recall falls by more than half. The F-score fails the same test for the opposite reason. It is genuinely size-neutral, and the enforced cases it recovers are the smaller ones—5.6% of small-firm cases against 1.9% of large-firm ones—which the restriction removes. Only the two networks discriminate on accounting content that survives holding size fixed.

Figure 4 shows the size bias directly. Among each model's within-year top-decile flagged firm-years, the median flagged firm has total assets of \$130 million under the Soft Label, well below the \$348 million population median, versus \$456 million under the F-score, \$863 million under the Hard Label, and \$1.56 billion under RUSBoost, with roughly two-thirds of RUSBoost's flags above the \$750 million cutoff. The enforcement-trained models concentrate their flags on large firms, reproducing the selection pattern of the enforcement process they are trained on; this is the empirical counterpart of the training problem described in the Introduction, and it motivates the size-controlled designs used in the validation tests below.

5 Validating the Hidden-Fraud Interpretation

The validation tests of this section are designed as a battery, not a list, because any single test can be passed by a model that captures something other than fraud. Each test therefore draws on a different source of information about hidden fraud—ex-post regulatory and audit scrutiny (Section 5.1), ex-ante creditor screening (Section 5.2), the firm's own financing behavior (Section 5.3), and market prices (Section 5.4)—and we require the conclusion "the soft-label model captures hidden fraud" to hold simultaneously across all four; no single alternative signal produces all four results jointly. Each subsection below states the economic mechanism behind its test before presenting the results. As the results will show, the Soft Label is the only one of the three machine-learning models that aligns on every test. This is not a deficiency of the alternatives: the F-score and RUSBoost were built (and our Hard Label CNN was trained) to agree with the enforcement label, an objective on which they perform exactly as designed; where a benchmark departs from the hidden-fraud predictions below, we trace the departure to its design—its training target, its size profile, or its inputs—rather than leave it unexplained.

The joint requirement also rules out the most plausible innocent reading of the flags: that the model simply identifies firms in financial distress or in ordinary business decline. That reading makes predictions of its own, and on two of the four tests they run opposite to ours: distressed firms face depressed share prices and limited access to equity markets, so a model that ranked them highest would flag firms that issue equity *less* often than the average firm, not more; and a portfolio formed on distress should earn little abnormal return once size, value, and profitability are held fixed, because those factors already absorb most of the compensation for distress risk.

5.1 Validation Against Post-Scrutiny Arthur Andersen Clients

Our first validation test asks whether each model's flags tilt toward or away from a population subjected to unusually intense *ex-post regulatory and audit scrutiny*. When Arthur Andersen collapsed in 2002, all of its corporate clients had to be re-audited by a successor auditor (predominantly the surviving Big 4), and a coordinated regulatory response and intense media spotlight focused on whether AA's prior audit work had concealed accounting irregularities. A non-AAER firm that survived this scrutiny is not certified fraud-free, but as a group the post-scrutiny AA cohort carries a lower residual hidden-fraud rate than the unscrutinized non-AA pool—a *relatively* clean anchor, and that weaker premise is all the test requires. The prediction is directional: a model that captures hidden fraud should tilt its high-risk flags *away* from this population, while a model that has merely inherited the size tilt of SEC enforcement will flag it *more*, because AA's former clients lean toward larger firms.

Table 5 implements the test on this post-scrutiny anchor. Both panels pool the Arthur Andersen client firm-years of 2002–2006 (identified by the 2001 audit signature, following Dyck et al., 2024), exclude AAER cases, and restrict to large firms with total assets above \$750 million—the canonical Dyck–Morse–Zingales sample—so that the anchor comparison is like-for-like on size and cannot be driven by the mechanical size tilt documented in Figure 4. The test is non-circular: the anchor is defined by the audit signature, not by any fraud label the models trained on.

Panel A is descriptive: within the large-firm pool, the 502 post-scrutiny AA firm-years are spread across each model's ranking roughly like the non-AA population, with RUSBoost placing the most in its top decile (60, or 12.0%) and the Soft Label the fewest (40, or 8.0%; Hard Label 46, or 9.2%). Panel B is the formal test: a logistic regression of the top-decile flag—the same within-year top decile of the run-averaged ranking used throughout the paper, computed inside the > \$750M pool—on the AA-client indicator, with firm controls, year fixed effects, size-decile×industry fixed effects, and firm-clustered standard errors; coefficients are log-odds. The Soft Label is the only one of the three models whose coefficient is significant at conventional levels: post-scrutiny AA clients' log-odds of being flagged are 0.65 lower than observably similar large firms' (-0.650, $z = -2.12$; the linear-probability estimate is -4.7 percentage points, $t = -2.20$), corresponding to a 2.4-percentage-point lower raw flag rate than the other large firms in the same pool (8.0% of AA firm-years reach the top decile, Panel A). The estimate is corroborated by randomization inference: across 300 firm-level reassignments of the AA label within industry×size strata, refitting the model on each draw, the observed coefficient is more negative than 98.3% of the placebo draws (one-sided $p = 0.017$). The Hard Label coefficient is smaller and insignificant (-0.336, $z = -1.11$), and RUSBoost's is positive and insignificant ($+0.159$, $z = 0.48$).

We emphasize what this test does and does not show: none of the three models actively over-flags the anchor once size is held fixed. The value of the test is directional and joint: the Soft Label is the only one of them that tilts its flags away at conventional (5%) significance from a cohort of firm-years subjected to unusually intense regulatory and audit scrutiny, which is what a genuine fraud signal (as opposed to an enforcement-salience signal) should do.

5.2 Bank-Loan Screening

The second validation test replaces ex-post regulatory scrutiny with *ex-ante creditor screening*, an entirely different source of information about hidden fraud. Banks accumulate private information through credit due diligence and continuing covenant monitoring—non-public reviews of customer–supplier relationships, internal control documents, and ongoing financial-statement reviews performed for covenant compliance—and they have strong financial incentives to identify accounting risk, because their own capital is at stake. The two mechanisms are complementary: one is ex-post and regulatory, the other ex-ante and contractual, so together they provide a much stronger negative benchmark than either could alone. If the Soft Label CNN captures accounting-quality patterns that such monitors screen on, firms with outstanding syndicated bank loans (which have passed bank due diligence and accept ongoing covenant monitoring) should be flagged less often than firms without such loans.

Table 6 implements this test. We build a firm-year indicator *has_loan* using DealScan linked to Compustat via the Chava–Roberts link, and restrict to non-AAER firms over 2000–2019.

Panel A reports the raw flag rates. Under the Soft Label, 5.4% of loan-holding firm-years fall in the within-year top decile of predicted fraud probability, versus 15.7% of firm-years without an active loan. Under every enforcement-trained benchmark the pattern reverses: loan holders are flagged *more* often than non-holders (11.9% vs. 7.7% for the Hard Label, 12.5% vs. 7.0% for RUSBoost, and 12.4% vs. 7.0% for the F-score), reflecting, for the Hard Label and RUSBoost, the concentration of bank borrowers among the larger firms where their flags concentrate (Figure 4). Panel B reports a multivariate logistic regression (log-odds coefficients) controlling for the Graham et al. (2008) bank-loan determinants (leverage, profitability, market-to-book, tangibility, the standardized Altman Z-score, and cash-flow volatility) plus year and size-decile×industry fixed effects. The Soft Label's *has_loan* coefficient is negative and significant (-0.541, $z = -7.20$; the linear-probability estimate in column (4) is -4.3 percentage points, $t = -8.07$). Once size deciles are held fixed, the Hard Label's coefficient is negative but insignificant (-0.104, $z = -1.42$) and RUSBoost's is negative and significant (-0.242, $z = -3.31$): their higher raw flag rates for borrowers in Panel A reflect size composition rather than borrower risk. What distinguishes the models is therefore the contrast between the two panels: the Soft Label avoids loan holders both unconditionally and within size–industry cells, whereas of the two benchmarks in Panel B only RUSBoost tilts away from them once size is held fixed—the Hard Label's coefficient is insignificant—so a screen built on their unconditional flags would over-sample bank-monitored firms.

In sum, firms that have passed bank due diligence, a classic ex-ante screening mechanism, are systematically avoided by the Soft Label's high-risk flags, in the raw rates and within size–industry cells alike. The size-controlled bank-loan coefficient alone does not separate the Soft Label from the benchmarks; the raw rates do, and so does the AA-client test. Combined with the AA-client analysis, this constitutes consistent cross-sectional evidence: firms that outside professional monitors independently screen (bank-monitored or post-scrutiny AA) are flagged less by the soft-label model, while the enforcement-trained alternatives flag bank borrowers at higher raw rates, and neither the Hard Label nor RUSBoost tilts significantly away from post-scrutiny AA clients. Which approach appears preferable therefore depends on the evaluation criterion: models built to agree with the enforcement label rank highest against that label, while externally screened benchmarks favor the model trained to look beyond it.

5.3 Future Equity Issuance

The two screening tests above are negative benchmarks: populations a genuine fraud signal should avoid. The third validation test provides the battery's *positive* behavioral signature, drawn from the firm's own financing behavior. The economic mechanism is well established in the seasoned-equity-offering literature: firms whose accounting overstates true performance face a closing window before the overstatement reverses, and they rationally exploit the temporary overvaluation by issuing equity (Loughran and Ritter, 1995; Teoh et al., 1998; Baker and Wurgler, 2002); once the accounting unravels through subsequent write-downs, earnings disappointments, or operational reality, the recently issued equity underperforms benchmark returns. This issue-and-underperform footprint is observable directly from market data and mechanically independent of SEC enforcement: a hidden fraudster who issues equity in $t+1$ and underperforms thereafter generates exactly the same observable footprint as an SEC-prosecuted fraudster would. If the Soft Label CNN identifies firms with unsustainable accounting performance, its top-decile firms should therefore be disproportionately likely to issue equity in the period immediately following the prediction, and the equity they issue should subsequently underperform; a detector trained on SEC labels alone, by contrast, has no direct mechanism through which to learn this issuance-timing signal.

5.3.1 Issuance Rates and Cross-Sectional Logit Evidence

Table 7 reports both raw issuance rates (Panel A) and logit estimates (Panel B) for the strictly future $[t+1, t+3]$ window, where issuance is a completed SDC-recorded seasoned equity offering (IPOs and convertibles excluded). In Panel A, Soft Label top-decile firms complete an SEO at a 27.1% rate over the next three years versus 9.0% in the bottom 90%, three times the base rate ($t = 29.11$). By contrast, Hard Label top-decile firms issue at the unconditional rate (10.8% vs. 10.8%, $t = -0.01$), RUSBoost top-decile firms issue *less* often than the bottom 90% (8.9% vs. 11.0%, $t = -5.22$), as expected given the large, mature firms its flags concentrate on (Figure 4), and the F-score shows a modest positive gap (12.8% vs. 10.3%, $t = 5.25$), consistent with its size-neutral accounting-ratio design carrying part of the issuance signal.

Panel B reports a logistic regression (log-odds coefficients) with Issue(t+1..t+3) as the dependent variable and a top-decile *High Risk* indicator as the regressor of interest, controlling for lagged equity issuance, leverage, ROA, Tobin's Q, momentum, book-to-market, and the seasoned-equity-offering determinants of DeAngelo et al. (2010) (cash holdings, firm age, R&D, idiosyncratic volatility, the Frank and Goyal (2003) financing deficit, and a dividend-payer dummy), plus year and size-decile×industry fixed effects. The High Risk coefficient is +0.397 (z = 5.67) under the Soft Label and +0.230 (z = 2.92) under the Hard Label—a roughly 49% versus 26% increase in the odds of a completed SEO—and statistically zero (z = 1.06) under RUSBoost. The lagged-issuance control (which loads at $\beta \approx 0.9$, $z \approx 16$) absorbs persistent-issuer behavior, so the coefficients are interpreted as the *newly emerging* component of issuance risk: the Soft Label's coefficient is roughly 1.7 times the Hard Label's, and RUSBoost shows no incremental issuance association once size, industry, and the issuance determinants are absorbed. The proportional-odds scale compresses the Soft Label's advantage relative to Panel A because its flags sit in cells with high baseline issuance propensity; the linear-probability estimates in columns (4)–(6) put the conditional gap at +6.1 percentage points for the Soft Label, +2.0 for the Hard Label, and +0.3 (insignificant) for RUSBoost.

5.3.2 Post-Issuance Returns

Table 8 reports post-issuance Fama–French 25-adjusted stock returns for firms that are both in the top decile of each model's predicted probability *and* issue equity. Under the Soft Label, top-decile equity issuers deliver economically large negative annualized FF25-adjusted returns over the post-issuance $[t{+}1, t{+}5]$ window: −11.9% per year on the value-weighted FF25 benchmark and −11.6% on the equal-weighted benchmark, both significant at the 1% level. The corresponding equal-weighted averages are −2.3% for Hard Label issuers and −3.5% for F-score issuers (a third and a half of the −6.9% earned by the average issuer in the full sample), and −0.2% (indistinguishable from zero) for RUSBoost issuers. Restricting to "Soft-only" firms (top-decile under Soft Label but not under Hard Label) sharpens the result to −13.3% (VW) and −13.0% (EW) over the same window.

The combined picture from Tables 7–8 is consistent with the soft-label design: the top-ranked Soft Label firms behave like firms with unsustainable accounting performance who opportunistically time equity issuance and then subsequently underperform the FF25 benchmark by economically large amounts, even though only a small fraction of them are ever subject to SEC enforcement. Among the benchmarks, it is the size-neutral F-score, not the enforcement-trained learners, that shows a partial version of this footprint (a positive issuance gap and −3.5% post-issuance returns), which reinforces the interpretation: the issue-and-underperform signature is carried by accounting-based fraud signals, and the Soft Label's contribution is to sharpen that signal severalfold rather than to introduce a pattern absent from the accounting literature.

5.4 Economic Significance: Long-Short Portfolio Alphas

The final test of the battery moves from sub-populations to market prices: it is the only test that integrates over the full cross-section rather than focusing on a particular group of firms, the only one that uses stock returns rather than accounting variables, and the only one that requires the model's predictions to be informative *after* adjusting for priced risk. The economic mechanism connecting hidden fraud to portfolio returns is gradual price adjustment: as fraud-related information slowly diffuses into prices through earnings disappointments, write-downs, and management changes, the firm's stock underperforms benchmark returns. If the Soft Label CNN successfully identifies firms with unsustainable accounting performance (both detected and undetected), then a portfolio that shorts the top-decile high-risk firms and holds the remaining 90% should produce positive returns as those firms' borrowed prosperity reverses. The Fama–French factor adjustment is essential to the test: it strips out returns attributable to standard risk premia (market, size, value, momentum, profitability, investment), so that any remaining alpha must come from fraud-related information that the factor model does not already capture.

Table 9 reports Fama–French factor regressions of monthly long-short portfolio returns for the three machine-learning methods: Soft Label (Panel A), Hard Label (Panel B) and RUSBoost (Panel C). Each year firms are independently ranked by each method and the top 10% are shorted while the remaining 90% are held long, equal-weighted, with a three-year overlapping holding window (signal years 2000–2019) under the Fama–French timing convention: a signal from the fiscal year ending in calendar year T enters the portfolio at the end of June of $T+1$, so every constituent's annual report is public before it is traded on. Constituent monthly returns are delisting-adjusted and winsorized at the 1st/99th percentiles before portfolio formation. AAER firms are excluded from the portfolios. Alphas are annualized and reported in percent; t-statistics use Newey–West standard errors with 12 lags, reflecting the serial correlation induced by the overlapping holding period.

The Soft Label portfolio earns an annualized three-factor alpha of 13.9% (t = 5.47) and a four-factor alpha of 12.9% (t = 5.55). The five-factor specification absorbs a large positive loading on RMW of 0.82 (t = 11.03), consistent with the strategy shorting lower-profitability firms, and the alpha declines to 8.4% but remains significant at the 1% level (t = 3.64). In contrast, none of the alternatives earns alpha significant at the 5% level in any specification: the Hard Label's three-factor alpha is −0.0% (t = −0.03), and RUSBoost's is −0.8% (t = −0.63). Across all specifications, the Soft Label alpha exceeds both alternatives by roughly 9 to 15 percentage points per year.

This pattern, comparable SEC-label detection rates but sharply different portfolio alphas, completes the four-test battery and directly supports the soft-label design. Hard Label and Soft Label both capture known AAER cases at similar rates (Table 4), but only Soft Label's predictions carry risk-adjusted return information, consistent with the soft label capturing accounting anomaly patterns that translate into future underperformance irrespective of whether the SEC subsequently prosecutes.

6 Mechanisms and Additional Analyses

6.1 Ablating the Training Target: Anchor, Tilt, and Their Combination

The validation evidence admits one remaining alternative interpretation, which we state in its strongest form. The soft label is constructed from future accounting outcomes, so perhaps the model has learned nothing about fraud at all: it may simply predict *which firms are about to deteriorate*, and frequent equity issuance, poor subsequent returns, and avoidance by externally screened populations might all be generic traits of firms headed downhill rather than signatures of concealed misreporting. If that reading were correct, the paper's central claim—that the soft-label model recovers hidden fraud—would reduce to a deterioration forecast.

We evaluate this interpretation by constructing the model it implies. Table 10 decomposes the soft-label target into its two components and evaluates each in isolation. The *de-anchored* network is the paper's own CNN—identical architecture, inputs, hyperparameter configuration, expanding windows, leakage rule and 30-run ensemble—retrained on a target from which all enforcement information has been removed: AAER firm-years receive the same accounting count label as every other firm-year, the fraud-replication step is dropped, and the label-neutralization rule is applied to every window. What remains is exactly the forward tilt, so this network is the strongest feasible implementation of a pure deterioration predictor within our framework. At the opposite vertex sits the Hard Label CNN, which trains on the enforcement anchor alone.

The decomposition delivers a two-sided verdict. On the detection side, the tilt alone is essentially blind to enforced fraud: at the top-1% cutoff the de-anchored network recovers 0.7% of AAER firm-years, against 18.3% for the Hard Label and 14.5% for the Soft Label. The asymmetry is mechanical rather than incidental. Manipulated reports make the manipulating firm's current accounting look *healthier*, so a signal built from "current statements predict future decline" points away from fraudulent firms almost by construction. On the validation side, the tilt alone produces only a faint shadow of the Soft Label's footprint. Its excess seasoned-equity rate—how much more often its flagged firms complete an offering over the next three years than the rest of the pool—is +4.9 percentage points ($t = 9.64$), against +18.1 points ($t = 29.11$) for the Soft Label. Its flags tilt away from post-scrutiny Arthur Andersen clients by 1.4 percentage points against the Soft Label's 2.4, and away from bank borrowers by 2.4 points against 10.3. In the size-controlled logit specifications of Tables 5 and 6, its Arthur Andersen coefficient is insignificant (−0.342, $z = -1.12$) where the Soft Label's is significant (−0.650, $z = -2.12$), and its bank-loan coefficient, while significant, is roughly a third of the Soft Label's magnitude (−0.152, $z = -2.33$, against −0.541, $z = -7.20$).

Panel C completes the picture and supplies the constructive half of the argument. The de-anchored network's top-decile flags overlap the Soft Label's by 33.3%, and the two rankings correlate substantially (Spearman +0.58), so the forward tilt is genuinely present inside the soft-label model; the Soft Label's rank correlation with the Hard Label is only +0.07. Reading the three vertices together: the anchor alone detects enforced fraud but carries no hidden-fraud footprint—its flagged firms issue equity at the unconditional rate, and it tilts *toward* bank borrowers by 4.2 percentage points; the tilt alone carries a faint footprint but detects almost nothing; their combination detects enforced fraud *and* produces the full footprint, an excess issuance rate of +18.1 percentage points where the two components separately deliver −0.0 and +4.9. The hidden-fraud signature is therefore not the sum of its parts but an interaction: the anchor teaches the network what enforced fraud looks like in the accounting matrix, and the tilt redirects that representation toward firms outside the enforcement set whose accounting subsequently unravels. This is the paper's claim that the training target is a first-order modeling choice, demonstrated inside a single architecture, and it rules out the deterioration-forecast reading of the validation evidence: a model that only predicted decline would sit at the de-anchored vertex, and that vertex fails the tests the Soft Label passes.

6.2 Investor Sentiment and the Predicted-Fraud Rate

Our final validation exercise moves from firm-level outcomes to the time series and cross-industry structure of the model's flags, asking whether they respond to an economic determinant of fraud that lies entirely outside any fraud label: investor sentiment. Theory and evidence connect the two: Wang et al. (2010) show that fraud propensity rises with investor beliefs about industry prospects, and Baker and Wurgler (2006) show that sentiment moves the prices of hard-to-value, hard-to-arbitrage firms the most. A model that genuinely tracks fraud incidence should therefore flag more firms in high-sentiment periods, and the effect should concentrate in sentiment-sensitive industries.

Figure 5 provides the aggregate, time-series view. Each model's annual flag rate (under its fixed in-sample top-decile cutoff) is plotted against the macro-orthogonalized Baker and Wurgler (2006) sentiment index.[4] The Soft Label flag rate co-moves with sentiment ($r = +0.72$); the realized SEC AAER count is also positively associated, consistent with genuine accounting-fraud incidence rising in high-sentiment periods; the Hard Label aggregate moves against sentiment ($r = -0.45$) and the F-score aggregate shows no association ($r = +0.26$), while the RUSBoost aggregate also co-moves ($r = +0.60$)—a first sign that a single aggregate series cannot separate the models. With twenty annual observations, however, aggregate correlations are suggestive at best: they have limited power to discriminate among models, and a single time series offers no way to hold industry composition or business conditions fixed.

The theory's distinctive prediction is in any case cross-sectional: if sentiment drives fraud, flag rates should rise most in the industries whose firms are hardest to value (Baker and Wurgler, 2006). Table 11 therefore moves the test into an industry-year panel, where identification no longer relies on the aggregate series at all. Industry-year flag rates are regressed on the interaction of sentiment with each industry's sentiment-sensitivity expo-

[4] Baker and Wurgler (2006) orthogonalize each of the six underlying sentiment proxies to a set of macroeconomic conditions before forming the index: growth in industrial production; real growth in durable, nondurable, and services consumption; growth in employment; and an NBER recession indicator. The orthogonalized index therefore measures investor sentiment beyond contemporaneous business-cycle fundamentals, so the co-movement documented here is not a mechanical reflection of macroeconomic conditions.

sure; industry and year fixed effects absorb all main effects—the year effects subsume, in particular, any aggregate co-movement of the kind Figure 5 displays—and industry-year fundamentals enter as controls. This is the exposure-times-aggregate-shock design of Rajan and Zingales (1998). The exposure is built the same way in both column blocks (the industry share of firms in the market-wide top decile of a composite score, the overpriced tail in which Stambaugh et al. (2015) show sentiment effects concentrate), and the blocks differ only in the score: the left block uses the published SYY composite mispricing score itself, while the right block applies SYY's average-of-rankings aggregation to the five valuation-uncertainty characteristics of Hribar and McInnis (2012) (the two exposures correlate at 0.71 across industries). Every specification includes the industry business-conditions controls of the fraud-waves literature (Wang et al., 2010) (industry asset growth, sales growth, ROA, and the prior year's industry stock return), so that the interaction captures sentiment beyond real industry booms. The Soft Label is the only one of the three models in the table whose interaction is positive and significant at conventional (5%) levels. A one-standard-deviation increase in sentiment raises the flag rate of a one-standard-deviation-more-exposed industry by 0.94 percentage points under the published-score exposure ($\delta = +0.943$, $t = 2.95$) and by 1.44 percentage points under the characteristics-composite exposure ($\delta = +1.443$, $t = 7.05$). The Hard Label interactions are insignificant in both blocks (-0.164, $t = -0.92$; -0.380, $t = -1.68$). RUSBoost's interaction is insignificant under both exposures ($+0.11$, $t = 0.95$ under the characteristics composite and $+0.05$, $t = 0.49$ under the published score). Because sentiment is external to any fraud label and the exposures combine published ingredients with a published aggregation method, this is out-of-sample evidence: the soft-label model uniquely tracks the Baker and Wurgler (2006) prediction (sentiment-driven fraud surfaces most in the industries whose firms are hardest to value), consistent with the industry-beliefs fraud channel of Wang et al. (2010).

7 Conclusion

This paper addresses a fundamental challenge in fraud detection: enforcement records reveal only a small fraction of actual fraud—Dyck et al. (2024) estimate that at least 10% of large publicly traded firms engage in fraud and that only about one-third of it is ever detected—so AAER-based training data are contaminated at scale by *hidden fraud*. Identifying these hidden fraudsters is the central goal of the paper, which we frame as a problem of recovering latent accounting manipulation under selectively observed enforcement labels. We replace the uniform zero target for unprosecuted firms with a soft label built from forward-looking accounting outcomes: a weakly supervised device that uses future accounting realizations as partial evidence on the latent state behind earlier reports.

Out of sample over 2000–2019, the soft-label model gives up little in SEC-labeled fraud capture relative to its identically specified hard-label twin, while much of the benchmarks' apparent full-sample performance turns out to reflect firm size rather than fraud. The validation evidence, in contrast, separates the models sharply and points uniformly in one direction. Soft Label top-decile firms issue equity at nearly three times the base rate and subsequently underperform by economically large margins. The two externally screened populations—post-scrutiny Arthur Andersen clients and bank-monitored borrowers—are systematically avoided by the Soft Label's flags; neither the Hard Label nor RUSBoost shows a significant tilt away from the post-scrutiny AA clients, and none of the enforcement-trained alternatives avoids bank borrowers unconditionally. A long-short portfolio on its ranking earns significant Fama–French alpha where neither the Hard Label nor RUSBoost earns any. And the flag rate co-moves with investor sentiment precisely in hard-to-value industries, an economic determinant of fraud external to any label.

Because firms that committed undetected fraud are never revealed as such, no single test could establish that the soft-label model identifies them. The tests above are designed to align only if the model captures genuine hidden fraud: top-decile firms display the issue-and-underperform fraud footprint, two externally screened populations (post-scrutiny AA and bank-monitored) are avoided rather than over-flagged, the long-short portfolio earns positive Fama–French alpha, and the flags respond to sentiment where theory says fraud should. Taken together, this mutually supporting evidence (behavioral, ex-post regulatory, ex-ante creditor, stock-return, and macro-financial) suggests that the soft-label model's predictions reflect hidden-fraud capture rather than statistical artifacts.

Beyond the specific construction, the paper's broader message is a diagnosis. Once enforcement labels are recognized as an incomplete and selectively revealed proxy for fraud, agreement with those labels and capture of the underlying fraud process become separate objectives, and the choice of training target becomes a first-order modeling decision rather than a default. The soft label is one way—surely not the only way—to internalize the second objective; what our evidence establishes is that the two objectives genuinely diverge, that the divergence is empirically first-order, and that future accounting outcomes are informative enough to exploit it. We are therefore confident about the problem and deliberately modest about the solution: future accounting outcomes prove the past only weakly, and the discipline lies in using them accordingly.

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Figure 1: The Iceberg Problem: Two Distortions from Enforcement-Based Labels

Schematic of the paper's setting. SEC enforcement (AAER) reveals only the tip of corporate fraud: our 2000–2019 evaluation sample contains 493 AAER firm-years, while the Dyck et al. (2024) pervasiveness estimates imply that only about one-third of fraud is ever detected, so at least twice as many fraudulent firm-years sit hidden below the waterline among the ≈53,600 unprosecuted observations. This structure creates two distinct distortions. The *training problem*: a model trained on hard AAER labels learns the properties of the visible tip—what enforcement selects, most notably large firms—rather than of fraud itself. The *evaluation problem*: any benchmark that treats non-AAER firm-years as verified negatives charges a model a "false positive" precisely when it surfaces hidden fraud. The paper's soft-label training target addresses the first distortion; its validation designs (post-scrutiny anchors, forward outcomes) address the second.

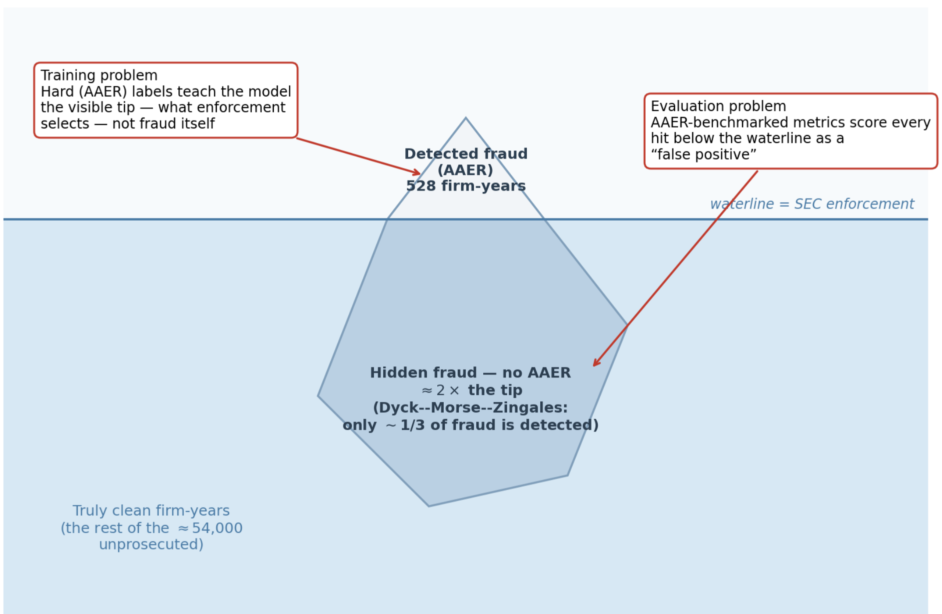

Figure 2: Soft-Label Construction Timeline: Where Future Information Can and Cannot Flow

Left: the information timeline for one test year τ. A training observation (i, t) feeds the model five years of inputs $[t-4, t]$ (blue); its soft label is built from the strictly forward window $\{t+1, t+2, t+3\}$ (orange), so future information enters the *training label only*, never the model inputs. (When the terminal-year label is unavailable—roughly a fifth of labeled non-AAER training windows, typically near data gaps or delisting—the most recent label within the window is used; most such fallbacks still draw on at least one post-t year, and in the small remainder (~5% of windows) the label is older, though all non-AAER labels stay within the narrow $[0.475, 0.525]$ band.) Training uses the expanding window $1990 \leq t \leq \tau-2$ (two-year gap), and the newest admitted cohorts ($t \in \{\tau-3, \tau-2\}$), whose label windows would touch the test year, have their non-AAER labels replaced by exactly 0.5 (gray), so no information dated τ or later influences any training label. The test prediction at τ uses inputs $[\tau-4, \tau]$ only. *Right*: the resulting label values. Non-AAER firm-years occupy a deliberately narrow band around 0.5 (from 0.475 for unflagged firm-years—73% of firm-years—to 0.525 for firm-years flagged on all four accounting indicators), while AAER firm-years keep the anchor label 1.0: a modest adjustment to the zero target, not a relabeling. *Bottom*: the training schedule across the full out-of-sample period, one row per test year $\tau = 2000, \ldots, 2019$: the model for τ trains on window-end cohorts $1990 \leq t \leq \tau-2$ (blue), with the two newest cohorts' non-AAER labels neutralized to 0.5 (gray), the gap year $\tau-1$ excluded, and predictions made at τ (red); the training window expands as τ advances.

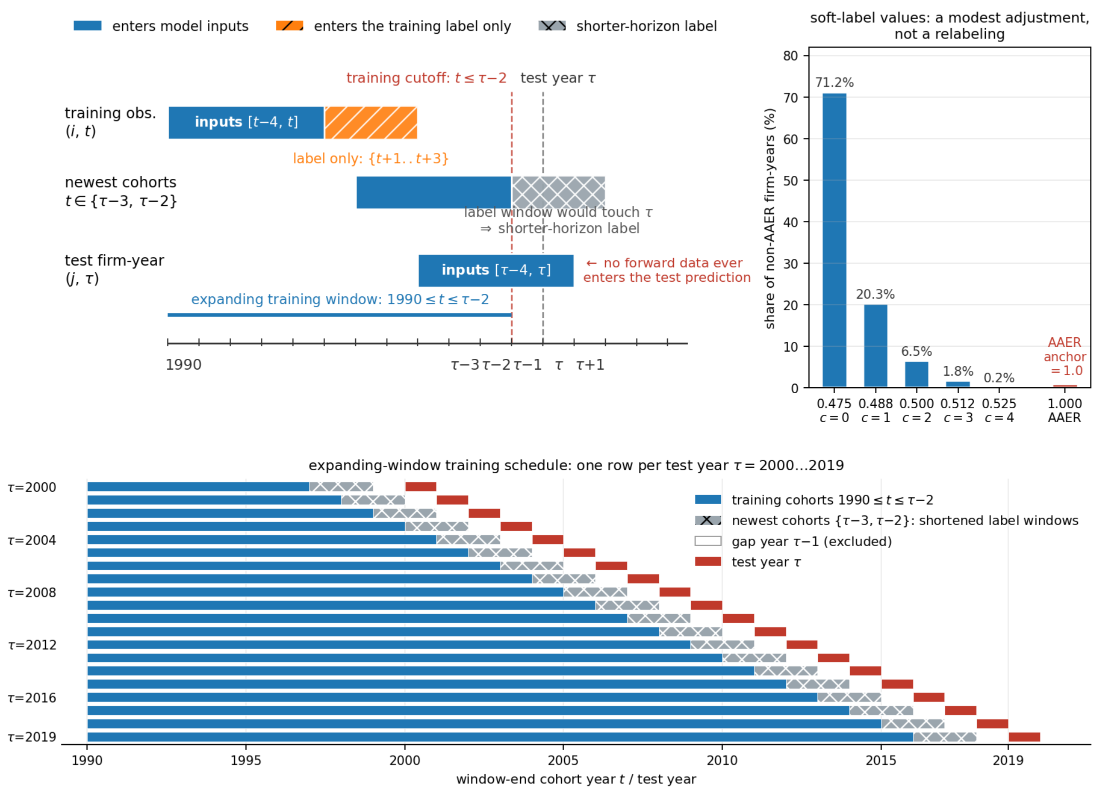

Figure 3: CNN-Based Fraud Detection Model Architecture with Residual Connections

The input is the 5×23 firm-year accounting matrix (five consecutive years of the 23 Compustat variables). Two convolutional blocks extract hierarchical features; the first includes a residual skip connection that facilitates gradient flow. The feature maps are flattened and passed through three fully connected layers to a sigmoid fraud probability. The certainty-weighted focal loss (inset) assigns higher weight to labels near 0 or 1 and lower weight to labels near 0.5, so the model learns jointly from hard AAER labels (anchored at 1.0) and the soft labels of non-AAER firm-years (a narrow band around 0.5).

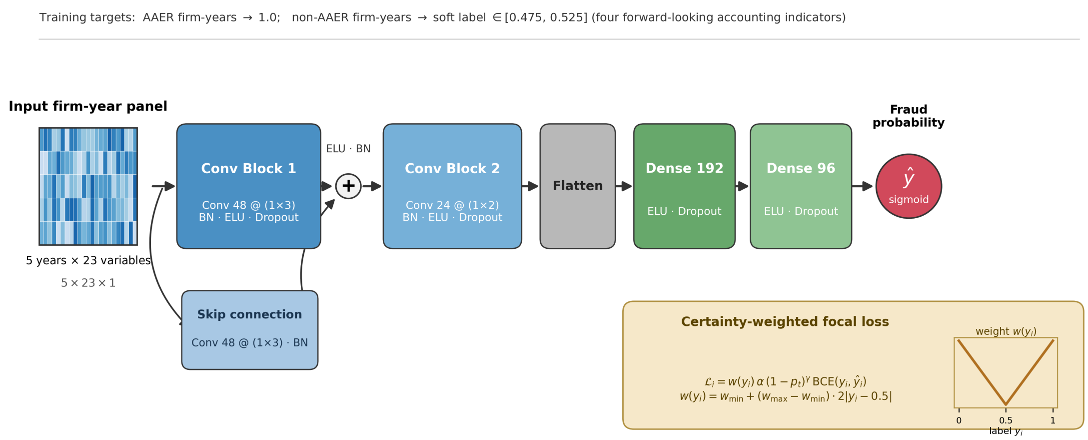

Figure 4: Firm-Size Distribution of Flagged Firms

Cumulative distribution of total assets among each model's within-year top-decile flagged firm-years (non-AAER, 2000–2019), against the distribution over all firm-years (dashed gray). The median flagged firm has total assets of $130M under the Soft Label, $456M under the F-score, $863M under the Hard Label, and $1.62B under RUSBoost, against $348M for the population; roughly two-thirds of RUSBoost's flags lie above the $750M Dyck–Morse–Zingales cutoff (dotted line). The enforcement-trained models concentrate their flags on large firms—the size bias inherited from the enforcement process, which motivates the size-controlled designs used throughout the paper—whereas the Soft Label's flags track (indeed slightly undershoot) the population size distribution.

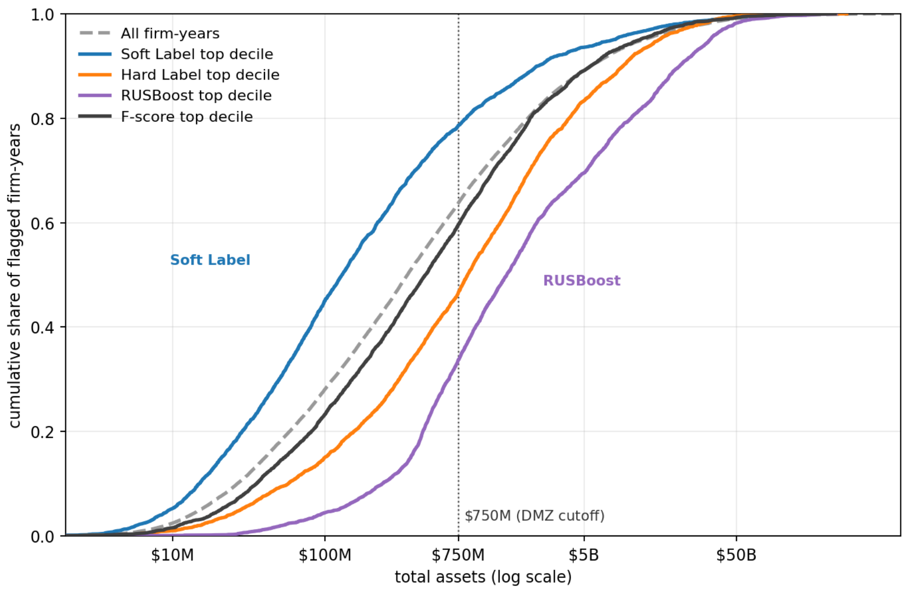

Figure 5: Baker–Wurgler Investor Sentiment and Aggregate Predicted Fraud Probability across Models

Annual aggregate fraud incidence versus the Baker and Wurgler (2006) investor-sentiment index orthogonalized to macroeconomic conditions (SENT_ORTH; dashed gray, right axis), 2000–2019. Panels A–D plot the annual fraction of non-AAER firms flagged by each model (left axis); Panel E plots the SEC AAER firm-year count as an external fraud-incidence reference. Firms are flagged against each model's *fixed in-sample* cutoff—the 90th percentile of its 2000-fold training-set scores, averaged over the 30 training runs—applied to every test year, so the threshold is calibrated out of sample; for the deterministic F-score the percentile is taken on Dechow's F-scale, whose raw probability scale drifts out of sample. Panel titles report the raw Pearson correlation with SENT_ORTH. The Soft-label flag rate co-moves with sentiment ($r = +0.72$), as does RUSBoost's ($r = +0.60$); the Hard-label aggregate moves against it ($r = -0.45$) and the F-score aggregate does not co-move ($r = +0.26$). The AAER count is also positively associated, though its post-2014 decline partly reflects enforcement-lag truncation (the AAER data end 2021); the model flag rates are unaffected.

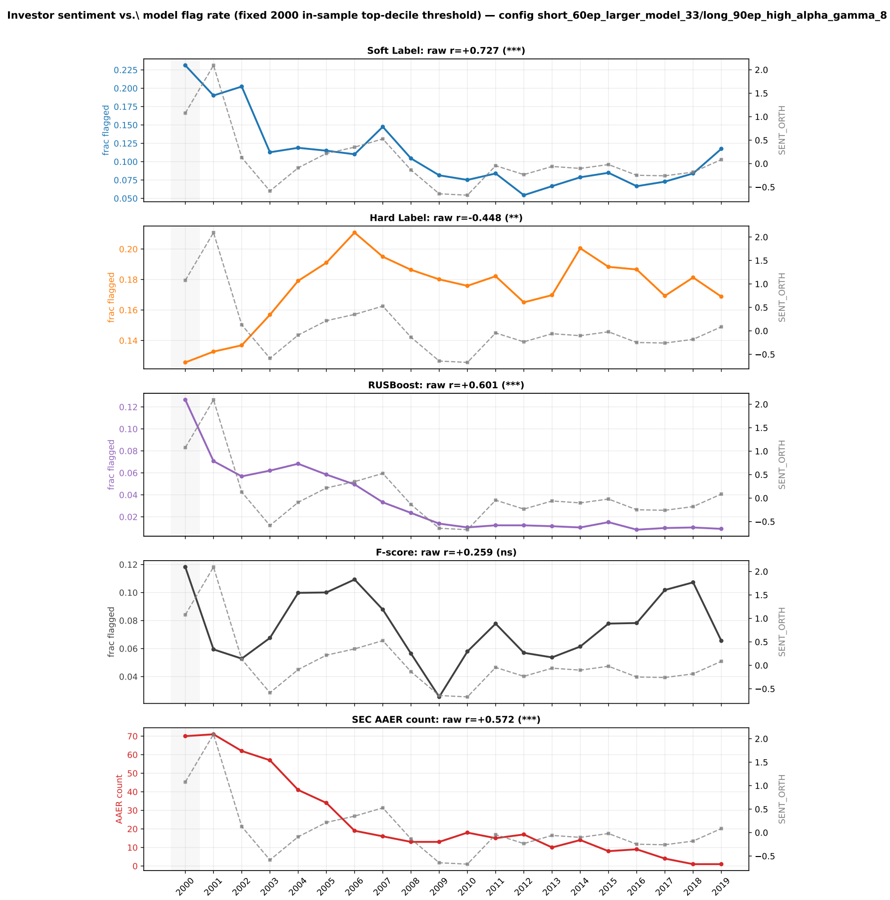

Table 1: Compustat Items and Variable Definitions

Panels A–D list the 23 raw Compustat (Annual) items used as model inputs; each firm-year requires non-missing values for all 23 items in all five rolling-window years (see Table 3). Panel E defines the main constructed variables and indicators. Ratios use Compustat Annual; returns from CRSP. AAER firms excluded from all analyses unless noted.

Variable	Definition
Panel A: Compustat Inputs — Assets (6 items)	
`che`	Cash and short-term investments
`rect`	Receivables, total
`invt`	Inventories, total
`act`	Current assets, total
`ppegt`	Property, plant, and equipment, gross
`at`	Total assets
Panel B: Compustat Inputs — Liabilities and equity (9 items)	
`ap`	Accounts payable, trade
`dlc`	Debt in current liabilities
`txp`	Income taxes payable
`lct`	Current liabilities, total
`dltt`	Long-term debt, total
`lt`	Total liabilities
`ceq`	Common/ordinary equity, total
`pstk`	Preferred/preference stock (capital, total)
`re`	Retained earnings
Panel C: Compustat Inputs — Income statement (6 items)	
`sale`	Sales/turnover, net
`cogs`	Cost of goods sold
`dp`	Depreciation and amortization
`txt`	Income taxes, total
`ib`	Income before extraordinary items
`ni`	Net income (loss)
Panel D: Compustat Inputs — Cash flow and market (2 items)	
`sstk`	Sale of common and preferred stock
`csho`	Common shares outstanding
Panel E: Constructed Variables and Indicators Used in Empirical Tables	
Leverage	(`dlc` + `dltt`) / `at`
ROA	`ni` / `at`
Tobin's Q	(CRSP December market equity + `lt`) / `at`
Momentum	Cumulative CRSP stock return over the $[t-12,\ t-2]$-month window (skipping the most recent month), measured at the fiscal year-end month
Book-to-Market	Book equity (Fama–French construction: `seq` + `txditc`− preferred stock, positive values only) / one-month-lagged market capitalization at the fiscal year-end month
Size decile	Within-fiscal-year decile of log(`at`)
Industry	Fama–French 48-industry classification (assigned from SIC)
Top 10% (High Risk)	$\mathbf{1}\{$within-year rank of predicted fraud probability $\geq 0.90\}$, computed separately under each scoring method (Soft Label, Hard Label, RUSBoost, F-score)
Within-year percentile rank	Empirical-CDF rank of predicted fraud probability within fiscal year, scaled to $[0, 100]$

Continued on next page

Table 1 – *Continued from previous page*

Variable	Definition
Issue$(t + k)$	$\mathbf{1}\{$firm completes an SDC-recorded seasoned equity offering (main tranche, non-withdrawn; IPOs and convertibles excluded) during its fiscal year $t{+}k\}$
Issue$(t{+}1..t{+}3)$	$\mathbf{1}\{$firm issues equity in any of fiscal years $\{t{+}1, t{+}2, t{+}3\}\}$
Lag Equity Issue	$\mathbf{1}\{$firm issued equity in fiscal year $t - 1\}$
AA Client	$\mathbf{1}\{$Compustat `au`=1 (Arthur Andersen) in fiscal 2001$\}$, following Dyck et al. (2024)
has_loan	$\mathbf{1}\{$firm has any active syndicated DealScan facility (start date ≤ FY-end t, maturity ≥ FY-start t)$\}$, linked to Compustat *gvkey* via the Chava–Roberts DealScan–Compustat link (2024-01 vintage); facility start and maturity dates from the WRDS DealScan facility table
FF25 adjusted return	Firm's annual raw stock return − return on the Fama–French 25 (size × book-to-market) benchmark portfolio to which the firm belongs

Table 2: Sample Selection

The Dechow et al. (2011) AAER filters (upper block; firms counted by company record) and the prediction-sample requirements (lower block; firms counted by distinct GVKEY—the first row collapses records to unique GVKEY firm-years: nine duplicate firm-year records, several company records sharing a GVKEY; the consecutive-years requirement drops no further events).

	Misstatement Firm-Years		Firms	
Selection Criterion	Remaining	Dropped	Remaining	Dropped
AAER database filters				
Annual file (earnings misstatement events)	2,195	—	942	—
Less: understatement cases	2,139	56	931	11
Less: missing GVKEY	1,755	384	744	187
Less: no Compustat match or missing total assets	1,619	136	702	42
Less: financial firms (SIC 6000–6999)	1,407	212	609	93
Prediction sample filters				
Less: collapse to unique GVKEY firm-years	1,398	9	597	12
Less: fiscal year outside 1990–2019 analysis window	1,193	205	492	105
Less: fewer than 5 years of prior data	1,012	181	407	85
Less: missing key financial variables	783	229	305	102
Final prediction sample	783		305	

Table 3: Sample Distribution by Year

This table presents the annual distribution of the sample, 1990–2019. **Compustat Universe:** all distinct Compustat firm-year records (unduplicated GVKEY-fiscal year), including financial firms. **23 Inputs Complete:** records with all 23 accounting inputs (Table 1) non-missing in that year. The window columns are taken directly from the training file used by all models. **Step 1 (Five-Year Window):** 135,897 firm-years with a complete, consecutive five-year history of the 23 Compustat inputs (Table 1); this requirement alone removes most financial firms, whose balance sheets lack several required working-capital items. **Step 2 (Training Sample):** windows with soft-label coverage, the panel all four models are trained on; the label construction (Section 3.1) requires a Compustat–CRSP match carrying market and book equity. This yields 84,089 firm-years (783 AAER). **Out-of-Sample Scored:** every 2000–2019 window the models score—those whose terminal year carries that same match—financial and utility firms included, 57,635 firm-years (501 AAER). Scoring needs only that terminal-year match, while labelling additionally needs a forward-looking count label that can be computed, so this column exceeds Step 2 in every year. The two are not nested: a window takes its label from the most recent of its years that carries one, so it can be labelled while its own terminal year has no match. The analysis then excludes financial (SIC 6000–6999) and utility (SIC 4900–4999) firms, 3,565 windows and 8 AAER firm-years, leaving the 54,070 firm-years (493 AAER) on which every model is ranked in Table 4. Oversampled training copies are excluded throughout.

Year	Compustat Universe	23 Inputs Complete	Step 1 Five-Year Window	Step 2 Training Sample	Out-of-Sample Scored	**AAER**
1990	9,571	6,155	4,023	2,401	—	9
1991	9,966	6,336	4,138	2,538	—	18
1992	10,705	6,736	4,205	2,657	—	20
1993	11,482	7,166	4,300	2,784	—	14
1994	11,899	7,581	4,405	2,884	—	13
1995	12,492	8,338	4,493	3,043	—	18
1996	12,624	8,534	4,704	3,280	—	24
1997	12,437	8,388	4,845	3,354	—	28
1998	12,554	8,675	4,948	3,409	—	42
1999	12,531	8,800	5,301	3,624	—	69
2000	12,093	8,473	5,191	3,556	3,724	74
2001	11,584	7,972	4,970	3,380	3,497	76
2002	11,253	7,657	5,148	3,475	3,604	64
2003	11,067	7,429	5,212	3,307	3,410	61
2004	10,901	7,288	5,206	3,179	3,325	46
2005	10,857	7,177	5,027	3,023	3,230	37
2006	10,878	7,001	4,863	2,902	3,103	22
2007	10,875	6,745	4,621	2,742	2,943	20
2008	10,687	6,529	4,441	2,667	2,868	14
2009	10,642	6,408	4,427	2,632	2,806	13
2010	10,860	6,335	4,349	2,548	2,708	18
2011	11,266	6,304	4,235	2,462	2,644	15
2012	11,669	6,695	4,159	2,438	2,645	18
2013	11,716	6,704	4,095	2,371	2,544	11
2014	11,550	6,455	4,069	2,316	2,508	14
2015	11,399	6,168	4,045	2,270	2,487	8
2016	11,367	5,925	4,236	2,269	2,447	9
2017	11,288	5,790	4,180	2,222	2,406	5
2018	11,372	5,716	4,092	2,190	2,380	2
2019	11,617	5,678	3,969	2,166	2,356	1
Total	341,202	211,158	135,897	84,089	57,635	783

Table 4: Detection Results by Year, 2000–2019: Soft Label vs. Hard Label vs. RUSBoost vs. Dechow F-score

Out-of-sample detection results for 2000–2019 on the aligned (Soft∩Hard∩RUS) sample, flagging the top 1% of firms within each year by predicted fraud probability (each learned model's within-year ranking averaged across its 30 training runs; the F-score is a single deterministic logit). Recall (detection rate) = TP/(TP+FN); precision = TP/flagged; because every method flags the same number of firms each year, the two move together. Total rows report pooled FN/TP counts, the year-by-year mean recall (panel years with no AAER firm-years—2018–2019 in Panel A—enter this mean as zero), and pooled precision. Panel A restricts to firms with total assets > $750M; Panel B uses the full aligned sample. The Dechow et al. (2011) F-score (Model 1) is re-estimated out of sample under the same expanding-window protocol and ranked within year where its inputs are non-missing.

Year	Soft Label		Hard Label		RUSBoost		F-score		Detection Rate (Recall)				Precision			
	FN	TP	FN	TP	FN	TP	FN	TP	Soft	Hard	RUS	F	Soft	Hard	RUS	F
Panel A: Firms with total assets > $750M																
2000	34	4	32	6	37	1	37	1	0.105	0.158	0.026	0.026	0.444	0.667	0.111	0.111
2001	37	4	37	4	39	2	39	2	0.098	0.098	0.049	0.049	0.444	0.444	0.222	0.222
2002	26	6	26	6	30	2	31	1	0.188	0.188	0.062	0.031	0.667	0.667	0.222	0.111
2003	22	6	21	7	24	4	28	0	0.214	0.250	0.143	0.000	0.600	0.700	0.400	0.000
2004	18	5	17	6	22	1	23	0	0.217	0.261	0.043	0.000	0.455	0.545	0.091	0.000
2005	14	5	14	5	17	2	19	0	0.263	0.263	0.105	0.000	0.455	0.455	0.182	0.000
2006	6	2	5	3	8	0	8	0	0.250	0.375	0.000	0.000	0.182	0.273	0.000	0.000
2007	7	1	7	1	8	0	8	0	0.125	0.125	0.000	0.000	0.091	0.091	0.000	0.000
2008	5	2	6	1	7	0	7	0	0.286	0.143	0.000	0.000	0.182	0.091	0.000	0.000
2009	5	2	5	2	7	0	7	0	0.286	0.286	0.000	0.000	0.182	0.182	0.000	0.000
2010	6	2	5	3	7	1	7	1	0.250	0.375	0.125	0.125	0.182	0.273	0.091	0.091
2011	5	3	4	4	8	0	8	0	0.375	0.500	0.000	0.000	0.273	0.364	0.000	0.000
2012	9	2	9	2	11	0	11	0	0.182	0.182	0.000	0.000	0.167	0.167	0.000	0.000
2013	6	2	6	2	8	0	8	0	0.250	0.250	0.000	0.000	0.167	0.167	0.000	0.000
2014	6	1	6	1	7	0	6	1	0.143	0.143	0.000	0.143	0.083	0.083	0.000	0.083
2015	3	0	2	1	3	0	3	0	0.000	0.333	0.000	0.000	0.000	0.083	0.000	0.000
2016	5	0	4	1	5	0	5	0	0.000	0.200	0.000	0.000	0.000	0.083	0.000	0.000
2017	1	0	1	0	1	0	1	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2018	0	0	0	0	0	0	0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2019	0	0	0	0	0	0	0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Total	215	47	207	55	249	13	256	6	0.162	0.206	0.028	0.019	0.213	0.249	0.059	0.027
Panel B: Full aligned sample																
2000	61	9	62	8	64	6	68	2	0.129	0.114	0.086	0.029	0.250	0.222	0.167	0.056
2001	61	10	61	10	66	5	69	2	0.141	0.141	0.070	0.028	0.294	0.294	0.147	0.059
2002	51	11	50	12	57	5	59	3	0.177	0.194	0.081	0.048	0.314	0.343	0.143	0.086
2003	46	11	47	10	51	6	56	1	0.193	0.175	0.105	0.018	0.333	0.303	0.182	0.030
2004	33	8	31	10	34	7	39	2	0.195	0.244	0.171	0.049	0.250	0.312	0.219	0.062
2005	26	8	27	7	29	5	33	1	0.235	0.206	0.147	0.029	0.258	0.226	0.161	0.032
2006	16	3	14	5	19	0	19	0	0.158	0.263	0.000	0.000	0.100	0.167	0.000	0.000
2007	14	2	15	1	16	0	15	1	0.125	0.062	0.000	0.062	0.071	0.036	0.000	0.036
2008	10	3	10	3	12	1	13	0	0.231	0.231	0.077	0.000	0.111	0.111	0.037	0.000
2009	9	4	9	4	12	1	13	0	0.308	0.308	0.077	0.000	0.148	0.148	0.037	0.000
2010	13	5	12	6	14	4	18	0	0.278	0.333	0.222	0.000	0.192	0.231	0.154	0.000
2011	11	4	10	5	15	0	14	1	0.267	0.333	0.000	0.067	0.160	0.200	0.000	0.040
2012	15	2	15	2	16	1	17	0	0.118	0.118	0.059	0.000	0.080	0.080	0.040	0.000
2013	9	1	8	2	9	1	10	0	0.100	0.200	0.100	0.000	0.042	0.083	0.042	0.000
2014	12	2	12	2	14	0	11	3	0.143	0.143	0.000	0.214	0.083	0.083	0.000	0.125
2015	8	0	7	1	8	0	7	1	0.000	0.125	0.000	0.125	0.000	0.042	0.000	0.043
2016	8	1	7	2	8	1	9	0	0.111	0.222	0.111	0.000	0.043	0.087	0.043	0.000
2017	4	0	3	1	4	0	4	0	0.000	0.250	0.000	0.000	0.000	0.043	0.000	0.000
2018	1	0	1	0	1	0	0	1	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.045
2019	1	0	1	0	1	0	1	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Total	409	84	402	91	450	43	475	18	0.145	0.183	0.065	0.083	0.152	0.165	0.078	0.033

Table 5: Validation Against Post-Scrutiny Arthur Andersen Clients: Soft Label vs. Hard Label vs. RUSBoost vs. Dechow F-score

An out-of-sample *specificity test* on a post-scrutiny anchor: the Arthur Andersen client firm-years of 2002–2006 (2001 AA audit signature, following Dyck et al., 2024), which emerged from the post-Enron re-audits with no AAER. These firms are not certified fraud-free; the scrutiny makes them *relatively* clean, which is all the test requires. Both panels pool 2002–2006, exclude AAER cases, and restrict to firms with total assets > $750M (the Dyck–Morse–Zingales sample), so the comparison is like-for-like on size; ranks and top-decile flags are computed within this pool, separately for each fiscal year. Panel A: where the AA firm-years sit in each model's ranking. Panel B: logit of the top-decile flag—the top decile of each model's run-averaged within-year ranking, the same flag construction used in every other table—on the AA-client indicator, with the controls shown, year and size-decile×industry fixed effects, and firm-clustered SEs; coefficients are log-odds, with z-statistics in parentheses. In the logit columns, fixed-effect cells in which the flag never varies are perfectly separated in the likelihood and are dropped, so N differs across models; the three right-hand columns re-estimate the same specification by OLS (linear probability model, firm-clustered SEs) on the full sample, where no cell is dropped and N is identical across models, with coefficients in units of the flag probability. The Soft Label estimate (a 2.4-percentage-point lower flag rate for AA clients than for the other large firms in the pool; 8.0% of AA firm-years reach the top decile, Panel A) is corroborated by randomization inference: 300 firm-level reassignments of the AA label within industry×size strata, refitting the model on each draw, give a one-sided $p = 0.017$. The soft-label network scores the full window universe here: its training routine had scored only the windows it also trained on, leaving 36,420 of 90,535 windows (40.2%) unscored, and those are recovered from the saved checkpoints without retraining. Financial (SIC 6000–6999) and utility (4900–4999) firms are excluded from the recovered windows—they are 1.45% of the training windows but 18.7% of the unscored ones, so admitting them would change the industry composition the network was estimated on rather than repair the scoring gap. $^{***}/^{**}/^{*}$: 1%/5%/10%.

	Soft Label	Hard Label	RUSBoost	F-score
Panel A: Distribution of Post-Scrutiny AA Firms across Probability Rankings				
In top 10%	8.0%	9.2%	12.0%	10.4%
# AA firm-years in top 10%	40 / 502	46 / 502	60 / 502	52 / 501

	Logit (log-odds)			OLS (linear probability)		
	Soft Label	Hard Label	RUSBoost	Soft Label	Hard Label	RUSBoost
Panel B: Specificity test — top-decile flag on the AA-client indicator						
AA-Client indicator	−0.650**	−0.336	+0.159	−0.047**	−0.031	+0.010
	(−2.12)	(−1.11)	(0.48)	(−2.20)	(−1.44)	(0.41)
Leverage	−0.484	+2.143***	+0.971	−0.054	+0.132***	+0.053
	(−0.75)	(3.00)	(1.19)	(−0.98)	(2.97)	(0.92)
ROA	−4.708***	+1.481	−2.433	−0.513***	+0.038	−0.257**
	(−4.02)	(1.36)	(−1.53)	(−4.49)	(0.94)	(−2.26)
Tobin's Q	−0.221**	−0.574***	+0.052	−0.016*	−0.022***	+0.006
	(−2.06)	(−3.68)	(0.49)	(−1.79)	(−4.06)	(0.59)
Momentum	+0.255	+0.005	+0.739***	+0.018	−0.001	+0.042***
	(1.35)	(0.03)	(3.38)	(1.23)	(−0.08)	(2.68)
Book-to-Market	−0.161	+0.054	−0.008	−0.016	+0.007	−0.003
	(−0.74)	(0.24)	(−0.04)	(−1.10)	(0.45)	(−0.19)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Size-decile × industry FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1,679	1,895	1,404	3,165	3,165	3,165
# AA firm-years	225	262	226	498	498	498

Table 6: Cross-Sectional Predicted Fraud Probability by Bank-Loan Status: Soft Label vs. Hard Label vs. RUSBoost vs. Dechow F-score

Predicted fraud probability by bank-loan status, 2000–2019, non-AAER firms. *has_loan*=1 if the firm has an active syndicated DealScan facility in fiscal year *t* (Chava–Roberts link; facility start and maturity dates from the WRDS DealScan facility table). Panel A reports the fraction of firm-years flagged in the within-year top decile, by loan status (all loan vs. no-loan differences significant at 1%). Panel B reports logistic regressions (log-odds coefficients) of the top-decile flag on *has_loan* with the Graham et al. (2008) bank-loan determinants, year and size-decile×industry fixed effects, and firm-clustered SEs; fixed-effect cells in which the flag never varies are perfectly separated in the likelihood and are dropped, so *N* differs across the logit columns; the three right-hand columns re-estimate the same specification by OLS (linear probability model, firm-clustered SEs) on the full sample, where *N* is identical across models. The Altman Z-score control is winsorized at 1/99 and standardized, so its coefficient is per one standard deviation. The F-score is the out-of-sample Dechow et al. (2011) Model 1. ***, **, *: 1%, 5%, 10%.

Panel A: Fraction flagged in within-year top 10%, by loan status (has_loan=1: $N = 29,632$; has_loan=0: $N = 23,945$)

	Loan	No loan
Soft Label	5.4%	15.7%
Hard Label	11.9%	7.7%
RUSBoost	12.5%	7.0%
F-score	12.4%	7.0%

Panel B: DV = $\mathbf{1}\{$firm-year in top-10% of within-year predicted fraud probability$\}$ — left: logit (log-odds); right: OLS (linear probability)

	Logit (log-odds)			OLS (linear probability)		
	Soft Label	Hard Label	RUSBoost	Soft Label	Hard Label	RUSBoost
	(1)	(2)	(3)	(4)	(5)	(6)
has_loan	−0.541***	−0.104	−0.242***	−0.043***	−0.006	−0.021***
	(−7.20)	(−1.42)	(−3.31)	(−8.07)	(−1.04)	(−3.79)
Leverage	−0.270	+0.178	+0.698***	−0.048***	+0.027*	+0.070***
	(−1.41)	(0.94)	(3.12)	(−3.46)	(1.70)	(4.11)
ROA	−1.994***	+0.739***	+0.041	−0.238***	+0.033***	+0.009*
	(−14.83)	(4.63)	(0.35)	(−14.71)	(5.04)	(1.71)
Tobin's Q	+0.126***	−0.036	+0.089***	+0.017***	−0.002	+0.003***
	(7.47)	(−1.32)	(4.39)	(8.43)	(−1.44)	(2.65)
Tangibility	−3.325***	−1.867***	−6.205***	−0.161***	−0.142***	−0.303***
	(−10.85)	(−7.10)	(−14.66)	(−12.27)	(−7.80)	(−17.16)
Z-score	−0.455***	−0.285***	−0.190***	−0.050***	−0.014***	−0.009***
	(−9.26)	(−4.25)	(−3.54)	(−10.57)	(−4.44)	(−3.31)
CF volatility	+0.115	+0.017	−0.043	+0.016	+0.001	−0.003***
	(1.36)	(0.43)	(−0.56)	(1.64)	(0.35)	(−4.18)
Pseudo R^2 / R^2 (Panel B)	0.274	0.091	0.264	0.246	0.075	0.205
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Size-decile × industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster SE (firm)	Yes	Yes	Yes	Yes	Yes	Yes
N	36,020	39,334	35,603	41,249	41,249	41,249

Table 7: Equity Issuance and Top-Decile Fraud Probability: Soft Label vs. Hard Label vs. RUSBoost vs. Dechow F-score

Panel A compares SDC seasoned-equity-offering rates (IPOs and convertibles excluded) between each model's top decile and bottom 90% on the non-AAER sample (t-statistics from two-sample tests). Panel B reports logistic regressions (log-odds coefficients) of Issue(t+1..t+3) on the top-decile *High Risk* indicator, controlling for lagged issuance, firm characteristics, and the DeAngelo et al. (2010) SEO determinants (including the Frank and Goyal, 2003 financing deficit), with year and size-decile×industry fixed effects and firm-clustered SEs; fixed-effect cells in which the DV never varies are perfectly separated in the likelihood and are dropped; the three right-hand columns re-estimate the same specification by OLS (linear probability model) on the full sample, where no cell is dropped and N is identical across models. The R&D control is winsorized at 1/99 and standardized, so its coefficient is per one standard deviation. The F-score is the out-of-sample Dechow et al. (2011) Model 1. ***, **, *: 1%, 5%, 10%.

Panel A: Equity Issuance Rates among Top-Decile Firms

	Soft Label		Hard Label		RUSBoost		F-score		t-statistic (Top–Bot)			
	Top 10%	Bot 90%	Top 10%	Bot 90%	Top 10%	Bot 90%	Top 10%	Bot 90%	Soft	Hard	RUS	F
Issue($t+1$)	15.9%	4.0%	5.2%	5.2%	4.8%	5.3%	6.5%	4.8%	+23.33***	+0.06	−1.69*	+4.81***
Issue($t+2$)	14.0%	3.8%	4.6%	4.9%	3.3%	5.0%	5.1%	4.6%	+21.16***	−0.90	−6.73***	+1.53
Issue($t+3$)	12.5%	3.4%	4.1%	4.3%	2.9%	4.5%	4.4%	4.1%	+19.77***	−0.94	−6.14***	+0.96
Issue(t+1..t+3)	27.1%	9.0%	10.8%	10.8%	9.2%	11.0%	12.8%	10.3%	+29.11***	−0.01	−4.41***	+5.25***
N (firm-years)	5,368	48,209	5,366	48,211	5,366	48,211	5,333	47,917				

Panel B: DV = Issue(t+1..t+3), IV = High Risk (Top Decile of Predicted Probability) — left: logit (log-odds); right: OLS (linear probability)

	Logit (log-odds)			OLS (linear probability)		
	Soft Label (1)	Hard Label (2)	RUSBoost (3)	Soft Label (4)	Hard Label (5)	RUSBoost (6)
High Risk	+0.397***	+0.230***	+0.087	+0.061***	+0.020**	+0.003
	(5.67)	(2.92)	(1.06)	(6.06)	(2.56)	(0.46)
Lag Equity Issue	+0.892***	+0.918***	+0.921***	+0.158***	+0.163***	+0.163***
	(15.52)	(16.03)	(16.09)	(16.09)	(16.47)	(16.53)
Leverage	+0.960***	+1.008***	+0.998***	+0.100***	+0.103***	+0.103***
	(6.17)	(6.48)	(6.44)	(5.86)	(6.04)	(6.02)
ROA	−0.231***	−0.318***	−0.315***	−0.050***	−0.064***	−0.063***
	(−3.14)	(−4.04)	(−4.01)	(−4.26)	(−5.26)	(−5.24)
Tobin's Q	+0.024*	+0.023*	+0.023*	+0.004**	+0.004**	+0.004**
	(1.90)	(1.85)	(1.81)	(2.12)	(2.12)	(2.07)
Momentum	+0.171***	+0.171***	+0.171***	+0.022***	+0.022***	+0.022***
	(7.97)	(7.90)	(7.90)	(7.36)	(7.29)	(7.31)
Book-to-Market	−0.054	−0.060	−0.058	−0.003	−0.003	−0.003
	(−1.26)	(−1.39)	(−1.34)	(−1.06)	(−1.22)	(−1.20)
Cash/Assets	−0.129	+0.021	+0.003	−0.001	+0.016	+0.014
	(−0.88)	(0.14)	(0.02)	(−0.05)	(0.94)	(0.84)

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Table 7 — Continued from previous page

Panel B: DV = Issue(t+1..t+3), IV = High Risk (Top Decile of Predicted Probability) — left: logit (log-odds); right: OLS (linear probability)

	Logit (log-odds)			OLS (linear probability)		
	Soft Label (1)	Hard Label (2)	RUSBoost (3)	Soft Label (4)	Hard Label (5)	RUSBoost (6)
Firm Age	−0.021***	−0.021***	−0.021***	−0.002***	−0.002***	−0.002***
	(−6.99)	(−7.06)	(−7.08)	(−7.09)	(−7.19)	(−7.24)
R&D	+0.087***	+0.090***	+0.090***	+0.023***	+0.024***	+0.023***
	(4.57)	(4.74)	(4.73)	(5.91)	(6.09)	(6.07)
Idio. Volatility	−0.119**	−0.104*	−0.106*	−0.014**	−0.013**	−0.013**
	(−2.04)	(−1.81)	(−1.84)	(−2.36)	(−2.14)	(−2.17)
Financing Deficit	+0.649***	+0.696***	+0.697***	+0.098***	+0.104***	+0.105***
	(6.25)	(6.65)	(6.66)	(7.31)	(7.66)	(7.70)
Dividend Payer	−0.648***	−0.647***	−0.645***	−0.043***	−0.043***	−0.043***
	(−7.40)	(−7.42)	(−7.38)	(−6.94)	(−6.96)	(−6.96)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Size-decile × Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
N	38,062	38,062	38,062	39,637	39,637	39,637
Pseudo R^2 / R^2	0.177	0.176	0.176	0.170	0.168	0.168

Table 8: Post-Issuance FF25 Adjusted Stock Returns: Soft Label vs. Hard Label vs. RUSBoost vs. Dechow F-score

Winsorized (1st/99th pct.) mean annual FF25-adjusted returns for equity issuers in each model's top decile. FF25 adjusted return = firm's annual raw return − matched Fama–French 25 (size×BM) portfolio return. Panel A uses value-weighted (VW) benchmark; Panel B uses equal-weighted (EW). "Soft-only" ("Hard-only"): top decile under Soft (Hard) only. Avg[1:5]: per-firm mean over the $[t{+}1, t{+}5]$ window. Stars from one-sample t-tests vs. zero. AAER firms excluded. ***, **, *: 1%, 5%, 10%.

	$t+1$	$t+2$	$t+3$	$t+4$	$t+5$	Avg[1:5]	N
Panel A: Value-Weighted (VW) FF25 Adjusted Returns							
Soft top-decile issuers	−12.4%***	−13.7%***	−11.6%***	−10.0%***	−6.2%**	−11.9%***	991
Soft-only (non-overlap)	−14.4%***	−14.8%***	−12.4%***	−13.1%***	−7.9%**	−13.3%***	900
Hard top-decile issuers	−1.0%	+0.0%	−5.6%**	−3.6%	−0.2%	−1.8%	347
Hard-only (non-overlap)	−3.7%	+1.6%	−5.9%*	−9.8%***	−2.4%	−3.2%*	256
RUSBoost top-decile issuers	−3.2%	−0.6%	+2.3%	+2.4%	+0.3%	+0.5%	531
F-score top-decile issuers	−4.3%*	−4.3%*	−4.6%*	−4.9%*	−1.0%	−3.1%**	443
Full sample issuers	−6.0%***	−7.3%***	−4.4%***	−6.9%***	−4.0%***	−6.5%***	3,139
Full sample non-issuers	+1.4%***	+1.4%***	+0.6%**	+0.6%**	+0.2%	+0.6%***	38,110
Panel B: Equal-Weighted (EW) FF25 Adjusted Returns							
Soft top-decile issuers	−13.4%***	−13.7%***	−10.8%***	−8.9%***	−5.1%*	−11.6%***	991
Soft-only (non-overlap)	−15.2%***	−14.8%***	−11.5%***	−11.8%***	−6.7%**	−13.0%***	900
Hard top-decile issuers	−3.0%	−1.0%	−5.7%**	−3.2%	−0.8%	−2.3%	347
Hard-only (non-overlap)	−5.6%*	+0.3%	−5.9%*	−9.3%***	−3.4%	−3.8%**	256
RUSBoost top-decile issuers	−5.0%**	−1.6%	+2.0%	+2.4%	+0.5%	−0.2%	531
F-score top-decile issuers	−5.6%**	−4.8%**	−4.7%*	−4.9%*	−1.3%	−3.5%**	443
Full sample issuers	−7.3%***	−7.8%***	−4.4%***	−6.6%***	−3.7%***	−6.9%***	3,139
Full sample non-issuers	−0.4%	+0.6%**	+0.3%	+0.3%	+0.0%	−0.2%	38,110

Table 9: Equal-Weighted Long-Short Portfolio Factor Model Tests: Soft Label vs. Hard Label vs. RUSBoost

Fama–French factor regressions of monthly long-short portfolio returns (bottom 90% − top 10%; AAER firms excluded; equal-weighted; annual rebalancing with a three-year overlapping holding period; signal years 2000–2019). Portfolios follow the Fama–French timing convention: a signal from the fiscal year ending in calendar year T enters at the end of June of T+1, so every constituent's annual report is public before it is traded on (at least a six-month lag for December fiscal-year ends). Constituent monthly returns are delisting-adjusted and winsorized at 1%/99% before portfolio formation. Alphas are annualized and in percent; t-statistics use Newey–West (12 lags) SEs. ***, **, *: 1%, 5%, 10%.

	α (%)	β_{MKT}	β_{SMB}	β_{HML}	β_{UMD}	β_{RMW}	β_{CMA}	R^2
Panel A: Soft Label								
3-Factor	13.859***	−0.140*	−0.423***	+0.489***				0.335
	(5.47)	(−1.75)	(−3.48)	(7.33)				
4-Factor	12.910***	−0.079	−0.414***	+0.532***	+0.143**			0.365
	(5.55)	(−1.26)	(−3.40)	(7.50)	(2.10)			
5-Factor	8.445***	−0.043	−0.189**	+0.429***		+0.815***	+0.040	0.561
	(3.64)	(−0.81)	(−2.35)	(8.44)		(11.03)	(0.38)	
Panel B: Hard Label								
3-Factor	−0.025	−0.082***	+0.087***	−0.025				0.145
	(−0.03)	(−3.87)	(2.93)	(−0.77)				
4-Factor	−0.222	−0.068***	+0.089***	−0.015	+0.033			0.163
	(−0.28)	(−3.76)	(3.15)	(−0.43)	(1.60)			
5-Factor	−0.389	−0.075***	+0.096***	−0.070***		+0.025	+0.069	0.160
	(−0.51)	(−4.12)	(3.09)	(−2.58)		(0.46)	(1.38)	
Panel C: RUSBoost								
3-Factor	−0.822	−0.155***	+0.155***	+0.095**				0.209
	(−0.63)	(−3.51)	(3.11)	(2.08)				
4-Factor	−1.284	−0.122***	+0.160***	+0.119**	+0.079*			0.250
	(−1.05)	(−4.26)	(3.39)	(2.33)	(1.73)			
5-Factor	−1.865	−0.137***	+0.202***	+0.033		+0.162	+0.027	0.254
	(−1.56)	(−4.63)	(3.68)	(0.93)		(1.38)	(0.30)	

Table 10: Decomposing the Training Target: Enforcement Anchor versus Forward Tilt

Ablation of the soft-label training target. The Hard Label CNN trains on the enforcement anchor alone (the binary AAER indicator); the de-anchored CNN trains on the forward tilt alone (AAER firm-years receive their own accounting count label, the fraud-replication step is dropped, and the label-neutralization rule is applied to every window); the Soft Label combines anchor and tilt. Of the 783 AAER windows, the 82 with no computable accounting count label leave the de-anchored training set—the same criterion every non-AAER window faces—and all of them are still scored out of sample. The de-anchored and the Soft networks share one pinned configuration, so the comparison between those two columns isolates the supervisory target, the replication and neutralization steps following mechanically from the absence of a positive class. The Hard Label carries the configuration the pre-committed 1999 validation rule selected for it, so its column differs in hyperparameters as well as in target: it is the paper's Hard Label benchmark rather than a like-for-like ablation of the anchor. Inputs, expanding windows, the leakage rule and the 30-run ensemble convention are identical across all three, and all three are evaluated on the pool of Table 4. Panel A reports yearly-mean AAER recall at the top-1% cutoff, the convention of Table 4; the Hard and Soft rates therefore coincide with that table's full-sample totals. Panel B reports each footprint measure as a single differential. *Excess SEO rate of flagged firms*: the three-year (t+1..t+3) SDC seasoned-equity-offering rate of the within-year top decile minus that of the bottom 90% (non-AAER pool; Welch t in parentheses), the rates of Table 7, Panel A. *Excess flag rate of post-scrutiny AA clients*: the top-decile flag rate of the Arthur Andersen clients of 2002–2006 minus that of other large firms (> $750M, non-AAER), on the estimation sample of Table 5, Panel B. *Excess flag rate of bank borrowers*: the flag rate of active syndicated-loan holders minus that of non-borrowers (2000–2019, non-AAER), the rates of Table 6, Panel A. Hidden-fraud capture predicts a positive issuance differential and negative AA-client and bank-loan differentials. Differentials are computed from unrounded rates and may differ from differences of the rounded levels by 0.1 pp. Panel C reports the share of the Soft Label's top-decile flags that each other network also flags, on the common non-AAER pool.

	Hard Label (anchor only)	De-anchored CNN (tilt only)	Soft Label (anchor + tilt)
Panel A: Detection of enforced fraud (yearly-mean AAER recall, top-1% cutoff)			
Top-1% detection rate	18.3%	0.7%	14.5%
Panel B: Hidden-fraud footprint (each row a differential)			
Excess SEO rate of flagged firms (pp)	−0.0	+4.9	+18.1
	(-0.01)	(9.64)	(29.11)
Excess flag rate of post-scrutiny AA clients (pp)	−1.2	−1.4	−2.4
Excess flag rate of bank borrowers (pp)	+4.2	−2.4	−10.3
Panel C: Relation to the Soft Label ranking			
Top-decile overlap with Soft flags	23.9%	33.3%	—

Table 11: Industry Sentiment Sensitivity and Predicted-Fraud Flag Rates

Industry-year panel, FF48 level, 2000–2019: $FlagRate_{i,t} = \delta\,(SENT_t \times E_i) + \boldsymbol{\theta}'\mathbf{W}_{i,t} + \alpha_i + \tau_t + u_{i,t}$, SEs clustered by industry. *FlagRate*: the percentage of non-AAER firms flagged under the model's fixed in-sample top-decile cutoff, in percentage points (so coefficients are in percentage points; t-statistics are invariant to the scaling); *SENT*: standardized macro-orthogonalized Baker and Wurgler (2006) index. The exposure E_i is the industry share of firms in the top decile of the published SYY mispricing score (Stambaugh and Yuan, 2017) (left block) or of an SYY-style composite of the five Hribar and McInnis (2012) valuation-uncertainty characteristics (right block), z-scored across industries. Controls $\mathbf{W}_{i,t}$: industry means of log size, leverage, book-to-market, retained earnings/assets, asset growth, sales growth, ROA, and the prior fiscal year's stock return, winsorized 1%/99%. The return control is lagged so that it is pre-determined relative to the flag rate it conditions: a return compounded over year t is realized over the same window as the year-t flags and, for non-December fiscal-year ends, extends past the prediction date. The interaction δ is unaffected by this choice. Coefficients on $\mathbf{W}_{i,t}$ are reported for completeness and are not the object of inference. RUSBoost's interaction is insignificant in both blocks (+0.11, t = 0.95 right; +0.05, t = 0.49 left). Section 6.2 details the construction. $^{*}/^{**}/^{***}$: 10%/5%/1%.

	Dependent variable: industry-year flag rate $FlagRate_{i,t}$ (FF48)					
	E_i: published SYY mispricing score			E_i: SYY composite of HM characteristics		
	Soft	Hard	RUS	Soft	Hard	RUS
$SENT_t \times E_i$	+0.943***	−0.164	+0.053	+1.443***	−0.380*	+0.113
	(2.95)	(−0.92)	(0.49)	(7.05)	(−1.68)	(0.95)
Mean log size	−0.819	+2.309	+0.745	−1.135	+2.449	+0.706
	(−0.52)	(1.28)	(0.95)	(−0.76)	(1.38)	(0.90)
Leverage	−2.108	+12.993***	−1.629	−0.089	+12.362***	−1.447
	(−0.37)	(3.23)	(−0.99)	(−0.02)	(3.19)	(−0.91)
Book-to-market	−1.707	+5.012***	+0.307	−1.696	+4.983***	+0.314
	(−0.73)	(3.04)	(0.30)	(−0.74)	(3.04)	(0.31)
Retained earnings	−0.065	+0.132	−0.079	+0.045	+0.100	−0.069
	(−0.23)	(0.47)	(−0.75)	(0.18)	(0.35)	(−0.67)
Asset growth	−0.518	+0.344	+1.533	+0.151	+0.072	+1.608
	(−0.22)	(0.16)	(1.05)	(0.07)	(0.03)	(1.08)
Sales growth	+4.392	+1.115	−0.839	+3.330	+1.454	−0.937
	(1.22)	(0.42)	(−0.81)	(1.04)	(0.55)	(−0.93)
ROA	−1.351	+5.239**	−2.055**	−0.811	+5.112**	−2.017**
	(−0.51)	(2.45)	(−2.55)	(−0.36)	(2.48)	(−2.58)
Prior-year return	−0.217	+0.618	+1.136	−0.217	+0.661	+1.126
	(−0.15)	(0.42)	(1.55)	(−0.17)	(0.45)	(1.54)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	515	515	515	515	515	515

Appendix A CNN Architecture, Loss Function, and Hyperparameter Configuration

This appendix provides the implementation details of the CNN architecture and loss function referenced in Section 3.1 and the Model Architecture subsection of the main text. As emphasized there, the architectural choices below are not the conceptual contribution of the paper, and the soft-label idea is in principle compatible with alternative architectures suitable for tabular time-series inputs.

Appendix A.1 Convolutional Network with Residual Connections

Convolutional Neural Networks (CNNs) automatically learn hierarchical feature representations through convolutional operations that capture spatial and temporal patterns (LeCun et al., 2015). We adapt the architecture to financial-statement data with residual connections (He et al., 2016) that facilitate gradient flow during training. The convolution operation applies a set of learnable filters across the input matrix:

$$Y_{i,j} = \sigma\left(\sum_{m}\sum_{n} W_{m,n} \cdot X_{i+m,j+n} + b\right), \tag{6}$$

where σ is the activation function and b is the bias. The network learns the filters W through backpropagation.

First convolutional block. The input passes through a convolutional layer with padding to maintain spatial dimensions. The kernel has height one in the year dimension and detects relationships among adjacent financial variables within the same time period (e.g., interactions among revenue, cost of goods sold, and gross profit, or among different expense categories); all cross-year interactions are formed by the fully connected layers, for the reasons discussed in the Model Architecture subsection of the main text. We apply batch normalization, an Exponential Linear Unit (ELU) activation

$$\mathrm{ELU}(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha(e^x - 1) & \text{if } x \leq 0, \end{cases} \tag{7}$$

with $\alpha = 1$, and dropout regularization. Unlike ReLU, ELU allows negative outputs and yields smoother gradients, which helps prevent the "dying ReLU" problem.

Residual connection. A skip connection passes the input through a separate convolutional layer to match dimensions and adds it to the output of the first block:

$$\mathrm{Output}_1 = \mathrm{BN}(\mathrm{ELU}(F_1(X) + G_1(X))), \tag{8}$$

where F_1 is the main convolutional path (Conv $\rightarrow$ BN $\rightarrow$ ELU $\rightarrow$ Dropout), G_1 is the residual path (Conv $\rightarrow$ BN), and BN denotes batch normalization. The skip connection alleviates the vanishing-gradient problem during backpropagation, allowing the network to learn incremental refinements rather than complete transformations.

Second convolutional block. The output of the first block passes through a second convolutional layer, followed by batch normalization, ELU activation, and dropout. This block captures pairwise interactions between the higher-level features produced by the first layer.

Fully connected layers and output. After the convolutional layers, the feature maps are flattened and passed through three fully connected (dense) layers, each with ELU activation and dropout. The final layer produces a scalar passed through a sigmoid activation to yield a fraud probability:

$$\hat{y} = \sigma(W_3 \cdot h_2 + b_3) = \frac{1}{1 + e^{-(W_3 \cdot h_2 + b_3)}}, \tag{9}$$

where h_2 is the output of the previous dense layer. Figure 3 illustrates the complete architecture.

Appendix A.2 Smooth Certainty-Weighted Focal Loss: Full Specification

The loss function in the main text builds on the Focal Loss framework of Lin et al. (2017), which addresses class imbalance by down-weighting well-classified examples and focusing on hard-to-classify cases. The full per-observation loss is

$$\mathcal{L}_i = w(y_i) \cdot \alpha\,(1 - p_t)^{\gamma} \cdot \mathrm{BCE}(y_i, \hat{y}_i), \tag{10}$$

where α is the balancing parameter and γ is the focusing parameter that reduces the relative loss for well-classified examples. The certainty weight $w(y_i)$ is

$$w(y_i) = w_{\min} + (w_{\max} - w_{\min}) \cdot \mathrm{certainty}(y_i), \qquad \mathrm{certainty}(y_i) = 2|y_i - 0.5|, \tag{11}$$

where $w_{\min}$ and $w_{\max}$ are tunable parameters controlling the weight range. The weight rises monotonically from $w_{\min}$ at $y_i = 0.5$ (maximum uncertainty) to $w_{\max}$ at $y_i \in \{0, 1\}$ (hard labels):

- When $y_i = 0$ or $y_i = 1$ (hard labels or very confident soft labels): $w_i = w_{\max}$.
- When $y_i = 0.5$ (maximum uncertainty): $w_i = w_{\min}$.

An advantage of this formulation is its smoothness and differentiability. Unlike threshold-based weighting schemes that can create gradient discontinuities, our symmetric certainty function produces stable gradients throughout training, facilitating effective optimization via backpropagation.

Appendix A.3 Optimization and Dropout

Optimization. We use the Adam optimizer with an adaptive learning rate that adjusts based on first and second moment estimates of gradients. The model is trained using mini-batch gradient descent. All random seeds for weight initialization, data shuffling, and dropout operations are fixed to ensure reproducibility and deterministic training.

Dropout regularization. Following Srivastava et al. (2014), we apply dropout after the first convolutional block, after the first dense layer, and after the second dense layer. Dropout works by randomly deactivating a subset of neurons during each training iteration. This stochastic process means that even when the same observation is presented to the network multiple times (as occurs under our fraud-replication augmentation), the model encounters it through different effective network architectures due to varying dropout patterns. Each replicated observation therefore contributes differently to the learning process, preventing the model from memorizing repeated instances. The network instead learns features that generalize across different subnetwork configurations, transforming simple data replication into genuine augmentation through multiple distinct dropout-induced "views" of each fraud case.

Appendix B Hyperparameter Grids and Configuration Selection

Appendix B.1 The configuration grid

Because neural-network results can be sensitive to training hyperparameters, we do not report a single hand-picked model. We train a grid of ten configurations, and we use the *same* ten configurations for both the Soft Label and the Hard Label networks, so that every Soft–Hard comparison in the paper holds the candidate set fixed. The grid (Table B1) spans the dimensions that matter most for this architecture: training length (60 to 200 epochs), learning rate (0.8 to 3.5×10^{-4}), network capacity (40/20 convolutional filters with 160/80 dense units, 48/24 with 192/96, or 56/28 with 224/112), dropout (0.080 to 0.100), and the focal-loss parameters ($\alpha \in [0.35, 0.50]$, $\gamma \in [2.8, 3.5]$); the batch size is held at 128 throughout. The certainty-weight range of the loss is held fixed at $[w_{\min}, w_{\max}] = [0.2, 5.0]$ throughout. Each configuration is trained as 30 independent runs (different random seeds for initialization, shuffling, and dropout), and all results in the paper use the 30-run-averaged predictions, so that no result depends on a single training draw. The RUSBoost benchmark is treated symmetrically: we estimate a ten-configuration, 30-runs-per-configuration grid that varies one hyperparameter at a time around the published Bao et al. (2020) baseline (Table B1, Panel B): the number of boosting iterations (the boosting analogue of training length), the learning rate, the minimum leaf size of the base decision trees, and the undersampling ratio.

Appendix B.2 The pre-committed selection rule

The paper's main tables report one pinned configuration per label. The selection was made once, by a rule fixed in advance, using only data that predates the 2000–2019 reporting window. The timeline is: training samples with window-ends through 1997; 1998 skipped by the two-year label-masking gap (Section 3.4.3); fiscal year 1999 as the validation fold; and 2000–2019 as the disjoint reporting window. We use the single 1999 fold rather than an average over earlier folds because the earlier folds are fraud-sparse (models trained through 1994 see only 74 fraud windows), so the validation signal in those folds is weak; 1999 is the most saturated fold and the closest to the reporting window.

Each label is selected on *its own training objective*. For the Hard Label, the criterion is the 1999 out-of-sample AUC against the AAER indicator, the binary objective the model is trained on. For the Soft Label, the criterion is the Spearman rank correlation between the run-averaged 1999 out-of-sample prediction and the soft label among *non-AAER* firms. This metric follows from the design in three steps: the graded non-AAER signal is the component of the Soft training target that is unique to it (the fraud anchor at 1.0 is shared with the Hard Label, and including AAER firms would turn the criterion into a recall proxy, the enforcement-based yardstick the paper argues is contaminated); the soft label is ordinal by construction (a monotone transform of the indicator count, with an arbitrary scale parameter), so a rank metric is the appropriate fit measure; and every downstream use of the model output is itself rank-based (within-year percentile ranks and top-decile flags). The criterion touches none of the paper's economic outcome variables, so the selection is non-circular with respect to every validation test.

Applying the rule across the ten configurations of Table B1 (Panel A reports each configuration's 1999 validation metric) selects configuration 1 for the Soft Label and configuration 2 for the Hard Label. The same rule is applied to RUSBoost. Its ten configurations are estimated on the same 1999 fold and judged on the same criterion as the Hard Label—1999 out-of-sample AUC against the AAER indicator—because RUSBoost is likewise trained on that binary label. The rule selects the minimum-leaf-10 configuration (Panel B, configuration 6: 300 boosting iterations, learning rate 0.1, minimum leaf size 10), whose 1999 AUC is 0.698 against 0.686 for the published Bao et al. (2020) configuration and which wins 89% of run-level bootstrap resamples; that is the configuration the paper reports. We apply the rule without exception rather than letting the benchmark inherit its published setting, so that no model in the comparison is chosen on a standard the others are not. The choice is not material. Estimated at the published configuration instead, RUSBoost's yearly-mean top-1% recall is 6.1% rather than 6.5% and its five-factor long-short alpha is -1.5% ($t = -1.25$) rather than -1.9% ($t = -1.56$); under either configuration it flags bank borrowers more often than non-borrowers, flags future equity issuers less often than the bottom 90%, shows no significant Arthur Andersen tilt, and earns no alpha significant at conventional levels in any factor specification.

Table B1: The Hyperparameter Configuration Grids

Panel A: the ten CNN training configurations estimated for both the Soft Label and Hard Label networks, ordered by training length. Conv C_1/C_2: filters in the two convolutional blocks; Dense D_1/D_2: units in the first two fully connected layers; LR in units of 10^{-4}; the same dropout rate is applied at the three dropout sites; α and γ are the focal-loss parameters (certainty-weight range fixed at $[0.2, 5.0]$). Soft ρ: Spearman correlation between the 30-run-averaged 1999 out-of-sample prediction and the soft label among non-AAER firms, the Soft Label selection criterion; Hard AUC: 1999 out-of-sample AUC against the AAER indicator, the Hard Label criterion. The pre-committed validation rule described above selects configuration 1 for the Soft Label and configuration 2 for the Hard Label. Panel B: the ten RUSBoost configurations, ordered by the number of boosting iterations, varying one hyperparameter at a time around the published Bao et al. (2020) baseline (configuration 3, the pinned benchmark): the number of boosting iterations, the learning rate, the minimum leaf size of the base decision trees, and the undersampling ratio (majority-to-minority ratio after random undersampling; 1.0 = balanced classes). Every configuration in both panels is trained as 30 independent runs, and all reported results use run-averaged predictions.

Panel A: CNN configurations (Soft Label and Hard Label)

ID	Epochs	Batch	LR (10^{-4})	Conv C_1/C_2	Dense D_1/D_2	Dropout	α	γ	Soft ρ	Hard AUC
1	60	128	3.5	48/24	192/96	0.080	0.35	3.0	0.226	0.697
2	90	128	2.0	40/20	160/80	0.080	0.42	3.2	0.218	0.712
3	90	128	2.0	48/24	192/96	0.080	0.38	3.0	0.208	0.705
4	100	128	2.0	40/20	160/80	0.080	0.35	2.8	0.217	0.708
5	100	128	2.0	40/20	160/80	0.080	0.35	3.5	0.217	0.697
6	120	128	1.0	40/20	160/80	0.085	0.38	3.0	0.214	0.705
7	120	128	2.0	40/20	160/80	0.080	0.35	2.8	0.212	0.708
8	150	128	1.0	56/28	224/112	0.100	0.42	3.2	0.199	0.697
9	150	128	1.5	40/20	160/80	0.085	0.38	3.0	0.201	0.698
10	200	128	0.8	56/28	224/112	0.100	0.50	3.5	0.186	0.702

Panel B: RUSBoost configurations

ID	Boosting iterations	Learning rate	Min. leaf size	Undersampling ratio
1	100	0.10	5	1.0
2	200	0.10	5	1.0
3	300	0.10	5	1.0
4	300	0.05	5	1.0
5	300	0.20	5	1.0
6	300	0.10	10	1.0
7	300	0.10	20	1.0
8	300	0.10	5	0.5
9	500	0.10	5	1.0
10	1,000	0.10	5	1.0

Appendix C Detection Results over Alternative Sample Windows

This appendix reproduces the by-year detection table (Table 4; firms with total assets > \$750M in Panel A, full aligned sample in Panel B) for the shorter out-of-sample sub-windows 2000–2008 (Table C1) and 2000–2014 (Table C2).

Table C1: Detection Results by Year, 2000–2008: Soft Label vs. Hard Label vs. RUS-Boost vs. Dechow F-score

Out-of-sample detection results for 2000–2008 on the aligned sample, reproducing Table 4 for the shorter sub-window; definitions and construction as there. Panel A: firms with total assets > \$750M; Panel B: full aligned sample.

Year	Soft Label		Hard Label		RUSBoost		F-score		Detection Rate (Recall)				Precision			
	FN	**TP**	**FN**	**TP**	**FN**	**TP**	**FN**	**TP**	**Soft**	**Hard**	**RUS**	**F**	**Soft**	**Hard**	**RUS**	**F**
Panel A: Firms with total assets > \$750M																
2000	34	4	32	6	37	1	37	1	0.105	0.158	0.026	0.026	0.444	0.667	0.111	0.111
2001	37	4	37	4	39	2	39	2	0.098	0.098	0.049	0.049	0.444	0.444	0.222	0.222
2002	26	6	26	6	30	2	31	1	0.188	0.188	0.062	0.031	0.667	0.667	0.222	0.111
2003	22	6	21	7	24	4	28	0	0.214	0.250	0.143	0.000	0.600	0.700	0.400	0.000
2004	18	5	17	6	22	1	23	0	0.217	0.261	0.043	0.000	0.455	0.545	0.091	0.000
2005	14	5	14	5	17	2	19	0	0.263	0.263	0.105	0.000	0.455	0.455	0.182	0.000
2006	6	2	5	3	8	0	8	0	0.250	0.375	0.000	0.000	0.182	0.273	0.000	0.000
2007	7	1	7	1	8	0	8	0	0.125	0.125	0.000	0.000	0.091	0.091	0.000	0.000
2008	5	2	6	1	7	0	7	0	0.286	0.143	0.000	0.000	0.182	0.091	0.000	0.000
Total	169	35	165	39	192	12	200	4	0.194	0.207	0.048	0.012	0.380	0.424	0.130	0.043
Panel B: Full aligned sample																
2000	61	9	62	8	64	6	68	2	0.129	0.114	0.086	0.029	0.250	0.222	0.167	0.056
2001	61	10	61	10	66	5	69	2	0.141	0.141	0.070	0.028	0.294	0.294	0.147	0.059
2002	51	11	50	12	57	5	59	3	0.177	0.194	0.081	0.048	0.314	0.343	0.143	0.086
2003	46	11	47	10	51	6	56	1	0.193	0.175	0.105	0.018	0.333	0.303	0.182	0.030
2004	33	8	31	10	34	7	39	2	0.195	0.244	0.171	0.049	0.250	0.312	0.219	0.062
2005	26	8	27	7	29	5	33	1	0.235	0.206	0.147	0.029	0.258	0.226	0.161	0.032
2006	16	3	14	5	19	0	19	0	0.158	0.263	0.000	0.000	0.100	0.167	0.000	0.000
2007	14	2	15	1	16	0	15	1	0.125	0.062	0.000	0.062	0.071	0.036	0.000	0.036
2008	10	3	10	3	12	1	13	0	0.231	0.231	0.077	0.000	0.111	0.111	0.037	0.000
Total	318	65	317	66	348	35	371	12	0.176	0.181	0.082	0.029	0.227	0.231	0.122	0.042

Table C2: Detection Results by Year, 2000–2014: Soft Label vs. Hard Label vs. RUS-Boost vs. Dechow F-score

Out-of-sample detection results for 2000–2014 on the aligned sample, reproducing Table 4 for the shorter sub-window; definitions and construction as there. Panel A: firms with total assets > $750M; Panel B: full aligned sample.

Year	Soft Label		Hard Label		RUSBoost		F-score		Detection Rate (Recall)				Precision			
	FN	TP	FN	TP	FN	TP	FN	TP	Soft	Hard	RUS	F	Soft	Hard	RUS	F
Panel A: Firms with total assets > $750M																
2000	34	4	32	6	37	1	37	1	0.105	0.158	0.026	0.026	0.444	0.667	0.111	0.111
2001	37	4	37	4	39	2	39	2	0.098	0.098	0.049	0.049	0.444	0.444	0.222	0.222
2002	26	6	26	6	30	2	31	1	0.188	0.188	0.062	0.031	0.667	0.667	0.222	0.111
2003	22	6	21	7	24	4	28	0	0.214	0.250	0.143	0.000	0.600	0.700	0.400	0.000
2004	18	5	17	6	22	1	23	0	0.217	0.261	0.043	0.000	0.455	0.545	0.091	0.000
2005	14	5	14	5	17	2	19	0	0.263	0.263	0.105	0.000	0.455	0.455	0.182	0.000
2006	6	2	5	3	8	0	8	0	0.250	0.375	0.000	0.000	0.182	0.273	0.000	0.000
2007	7	1	7	1	8	0	8	0	0.125	0.125	0.000	0.000	0.091	0.091	0.000	0.000
2008	5	2	6	1	7	0	7	0	0.286	0.143	0.000	0.000	0.182	0.091	0.000	0.000
2009	5	2	5	2	7	0	7	0	0.286	0.286	0.000	0.000	0.182	0.182	0.000	0.000
2010	6	2	5	3	7	1	7	1	0.250	0.375	0.125	0.125	0.182	0.273	0.091	0.091
2011	5	3	4	4	8	0	8	0	0.375	0.500	0.000	0.000	0.273	0.364	0.000	0.000
2012	9	2	9	2	11	0	11	0	0.182	0.182	0.000	0.000	0.167	0.167	0.000	0.000
2013	6	2	6	2	8	0	8	0	0.250	0.250	0.000	0.000	0.167	0.167	0.000	0.000
2014	6	1	6	1	7	0	6	1	0.143	0.143	0.000	0.143	0.083	0.083	0.000	0.083
Total	206	47	200	53	240	13	247	6	0.215	0.240	0.037	0.025	0.292	0.329	0.081	0.037
Panel B: Full aligned sample																
2000	61	9	62	8	64	6	68	2	0.129	0.114	0.086	0.029	0.250	0.222	0.167	0.056
2001	61	10	61	10	66	5	69	2	0.141	0.141	0.070	0.028	0.294	0.294	0.147	0.059
2002	51	11	50	12	57	5	59	3	0.177	0.194	0.081	0.048	0.314	0.343	0.143	0.086
2003	46	11	47	10	51	6	56	1	0.193	0.175	0.105	0.018	0.333	0.303	0.182	0.030
2004	33	8	31	10	34	7	39	2	0.195	0.244	0.171	0.049	0.250	0.312	0.219	0.062
2005	26	8	27	7	29	5	33	1	0.235	0.206	0.147	0.029	0.258	0.226	0.161	0.032
2006	16	3	14	5	19	0	19	0	0.158	0.263	0.000	0.000	0.100	0.167	0.000	0.000
2007	14	2	15	1	16	0	15	1	0.125	0.062	0.000	0.062	0.071	0.036	0.000	0.036
2008	10	3	10	3	12	1	13	0	0.231	0.231	0.077	0.000	0.111	0.111	0.037	0.000
2009	9	4	9	4	12	1	13	0	0.308	0.308	0.077	0.000	0.148	0.148	0.037	0.000
2010	13	5	12	6	14	4	18	0	0.278	0.333	0.222	0.000	0.192	0.231	0.154	0.000
2011	11	4	10	5	15	0	14	1	0.267	0.333	0.000	0.067	0.160	0.200	0.000	0.040
2012	15	2	15	2	16	1	17	0	0.118	0.118	0.059	0.000	0.080	0.080	0.040	0.000
2013	9	1	8	2	9	1	10	0	0.100	0.200	0.100	0.000	0.042	0.083	0.042	0.000
2014	12	2	12	2	14	0	11	3	0.143	0.143	0.000	0.214	0.083	0.083	0.000	0.125
Total	387	83	383	87	428	42	454	16	0.186	0.204	0.080	0.036	0.190	0.199	0.096	0.037